

#1 · Basic Sciences

What structures pass through the diaphragm at T8, T10, and T12?

Answer

The inferior vena cava passes at T8, the esophagus and vagus nerves at T10, and the aorta, azygos vein, and thoracic duct at T12.

#2 · Basic Sciences

Which nerve roots form the phrenic nerve, and why does this matter for interscalene blocks?

Answer

The phrenic nerve arises from C3, C4, and C5. An interscalene brachial plexus block reliably causes ipsilateral hemidiaphragmatic paresis because local anesthetic spreads to these adjacent roots.

#3 · Basic Sciences

What is the sensory innervation of the airway above and below the vocal cords?

Answer

The internal branch of the superior laryngeal nerve supplies sensation above the vocal cords, and the recurrent laryngeal nerve supplies sensation below the vocal cords; both are branches of the vagus nerve.

#4 · Basic Sciences

What is the consequence of unilateral versus bilateral recurrent laryngeal nerve injury?

Answer

Unilateral injury causes hoarseness from an immobile, paramedian vocal cord. Bilateral injury causes both cords to assume a paramedian position, producing stridor and potential airway obstruction requiring emergent intervention.

#5 · Basic Sciences

At what vertebral level does the spinal cord end in adults, and why is this clinically relevant?

Answer

The spinal cord (conus medullaris) ends at approximately L1 to L2 in adults. Needle placement for neuraxial anesthesia should stay at or below L2 to L3 to avoid direct spinal cord trauma.

#6 · Basic Sciences

Name the layers a spinal needle traverses from skin to subarachnoid space via the midline approach.

Answer

Skin, subcutaneous tissue, supraspinous ligament, interspinous ligament, ligamentum flavum, epidural space, dura mater, and arachnoid mater.

#7 · Basic Sciences

Which brachial plexus cords and terminal nerves lie around the axillary artery at the level of an axillary block?

Answer

The musculocutaneous nerve typically lies within or lateral to coracobrachialis, while the median, ulnar, and radial nerves surround the axillary artery, with the radial nerve usually posterior to the artery.

#8 · Basic Sciences

What landmark defines the boundary for a supraclavicular versus infraclavicular brachial plexus block?

Answer

The supraclavicular block targets the trunks and divisions of the plexus just above the clavicle near the subclavian artery, while the infraclavicular block targets the cords, which are named for their relationship to the axillary artery below the clavicle.

#9 · Basic Sciences

What is the surface landmark and depth relationship for identifying the epidural space at L3 to L4 in an average adult?

Answer

The L3 to L4 interspace lies at the level of the intercrystal (Tuffier's) line connecting the iliac crests; the ligamentum flavum to epidural space is typically reached at 4 to 6 cm from skin in an average-weight adult.

#10 · Basic Sciences

Which cranial nerve provides motor innervation to all extraocular muscles except two, and which two are excepted?

Answer

The oculomotor nerve (CN III) innervates all extraocular muscles except the lateral rectus, innervated by the abducens nerve (CN VI), and the superior oblique, innervated by the trochlear nerve (CN IV).

#11 · Basic Sciences

What anatomic feature makes the right mainstem bronchus more prone to accidental endobronchial intubation?

Answer

The right mainstem bronchus is shorter, wider, and takes off at a less acute angle from the trachea than the left mainstem bronchus, making an endotracheal tube more likely to advance into it.

#12 · Basic Sciences

Which structures make up the borders of the femoral triangle, and what is their relative position for femoral nerve blockade?

Answer

The femoral triangle is bordered by the inguinal ligament superiorly, sartorius laterally, and adductor longus medially. From lateral to medial the contents are the femoral nerve, artery, vein, and lymphatics (mnemonic NAVEL).

#13 · Basic Sciences

How does an oxygen analyzer (galvanic or polarographic fuel cell) function in an anesthesia machine?

Answer

It generates an electrical current proportional to the partial pressure of oxygen through an electrochemical reaction between electrodes bathed in an electrolyte gel, consuming oxygen as it measures it.

#14 · Basic Sciences

What physical principle allows a pulse oximeter to estimate arterial oxygen saturation?

Answer

It uses the differential absorption of two wavelengths of light, red (660 nm) and infrared (940 nm), by oxyhemoglobin versus deoxyhemoglobin, isolating the pulsatile arterial component from venous and tissue absorption.

#15 · Basic Sciences

Why does pulse oximetry become unreliable in the presence of methemoglobinemia?

Answer

Methemoglobin absorbs light nearly equally at both 660 nm and 940 nm, driving the calculated saturation toward approximately 85 percent regardless of true oxygenation status.

#16 · Basic Sciences

What is the physical basis of infrared capnography for measuring end-tidal carbon dioxide?

Answer

Carbon dioxide absorbs infrared light at a specific wavelength (around 4.26 micrometers), and the amount of absorption measured across a sample or in the gas stream is proportional to the CO₂ concentration.

#17 · Basic Sciences

Differentiate mainstream from sidestream capnography.

Answer

Mainstream capnography places the sensor directly in the breathing circuit at the airway, giving a fast response with no sampling delay but adding weight and dead space; sidestream capnography aspirates a gas sample through tubing to a remote analyzer, adding a time delay but keeping the airway connector lightweight.

#18 · Basic Sciences

What does a capnography waveform with a slow, sloping upstroke (rather than a sharp rise) suggest?

Answer

A slow upstroke suggests uneven emptying of alveolar units with varying time constants, classically seen in bronchospasm, COPD, or a partially kinked or obstructed expiratory limb.

#19 · Basic Sciences

What does a sudden, sustained decrease in end-tidal CO₂ to near zero most often indicate intraoperatively?

Answer

Loss of the capnography waveform to near zero most often indicates esophageal intubation, circuit disconnection, or complete airway obstruction; if the patient remains ventilated and connected, it may also signal cardiac arrest with loss of pulmonary blood flow.

#20 · Basic Sciences

How does a gradual, progressive decline in end-tidal CO₂ over several breaths differ in significance from a sudden drop?

Answer

A gradual decline over several breaths suggests decreasing pulmonary perfusion or cardiac output, such as with hemorrhage, pulmonary embolism, or worsening hypotension, rather than an acute mechanical event.

#21 · Basic Sciences

What is the working principle of a variable orifice (Thorpe tube) flowmeter?

Answer

Gas flow lifts a bobbin or ball within a tapered glass tube; as flow increases, the annular gap around the float widens, and the float's resting height is read against a calibrated scale to indicate flow rate.

#22 · Basic Sciences

Explain the pin index safety system and diameter index safety system, and what each is designed to prevent.

Answer

The pin index safety system uses a unique pin arrangement on the yoke to prevent attaching the wrong compressed gas cylinder to a hanger yoke. The diameter index safety system uses gas-specific fitting diameters on pipeline connections to prevent connecting the wrong pipeline gas source to the machine.

#23 · Basic Sciences

What safety feature prevents delivery of a hypoxic gas mixture in most modern anesthesia machines?

Answer

A proportioning system, such as the Link-25 mechanism, mechanically or pneumatically links the nitrous oxide and oxygen flow controls so that the fraction of oxygen delivered cannot fall below approximately 21 to 25 percent.

#24 · Basic Sciences

How does a vaporizer maintain a constant output concentration despite changes in fresh gas flow?

Answer

Modern variable-bypass vaporizers use a temperature-compensating bypass valve that automatically adjusts the ratio of gas flowing through the vaporizing chamber versus the bypass chamber, keeping output concentration constant as flow and temperature vary.

#25 · Basic Sciences

Why would tipping an older variable-bypass vaporizer cause a dangerous overdose?

Answer

Tipping can allow liquid anesthetic to spill into the bypass chamber or fresh gas outlet, dramatically increasing the delivered vapor concentration far above the dial setting.

#26 · Basic Sciences

What does soda lime absorb, and what is the chemical reaction that produces heat as a byproduct?

Answer

Soda lime absorbs exhaled carbon dioxide through a reaction in which CO_2 combines with water to form carbonic acid, which then reacts with the hydroxides to form carbonates and water, an exothermic reaction that warms and humidifies the circuit.

#27 · Basic Sciences

Why is desiccated (dried-out) soda lime associated with compound A and carbon monoxide production?

Answer

Desiccated soda lime, particularly with strong bases like barium or potassium hydroxide, promotes degradation of sevoflurane into compound A and degradation of volatile agents (especially desflurane) into carbon monoxide.

#28 · Basic Sciences

What is the mechanism of action of neuromuscular blocking agents at the neuromuscular junction, contrasting depolarizing and nondepolarizing agents?

Answer

Depolarizing agents like succinylcholine bind and activate the nicotinic acetylcholine receptor, causing sustained depolarization and resultant blockade, while nondepolarizing agents competitively antagonize the receptor without activating it, preventing acetylcholine from binding.

#29 · Basic Sciences

What is the mechanism by which sugammadex reverses rocuronium or vecuronium?

Answer

Sugammadex is a modified gamma-cyclodextrin that encapsulates rocuronium or vecuronium molecules directly in the plasma, forming a tight inclusion complex that lowers the free concentration of the neuromuscular blocker and reverses blockade regardless of acetylcholinesterase activity.

#30 · Basic Sciences

By what mechanism do volatile anesthetics primarily produce general anesthesia at the molecular level?

Answer

Volatile anesthetics are thought to act mainly by potentiating inhibitory ligand-gated ion channels, particularly GABA-A receptors, and inhibiting excitatory channels such as NMDA receptors, rather than by a single unifying mechanism.

#31 · Basic Sciences

What determines the speed of onset of a volatile anesthetic in terms of blood and tissue solubility?

Answer

Agents with low blood-gas solubility (such as desflurane and nitrous oxide) equilibrate quickly between alveolus and blood, producing faster onset and offset, while agents with higher blood-gas solubility (such as isoflurane) equilibrate more slowly.

#32 · Basic Sciences

How does propofol's context-sensitive half-time behave with prolonged infusion compared to a drug like fentanyl?

Answer

Propofol has a relatively short and only modestly increasing context-sensitive half-time even after prolonged infusion, allowing predictable, timely emergence, whereas fentanyl's context-sensitive half-time increases substantially with infusion duration due to accumulation in peripheral compartments.

#33 · Basic Sciences

What receptor mediates the analgesic and hemodynamic effects of dexmedetomidine?

Answer

Dexmedetomidine is a selective alpha-2 adrenergic agonist that produces sedation, analgesia, and sympatholysis, causing characteristic bradycardia and hypotension by decreasing central sympathetic outflow.

#34 · Basic Sciences

Why does succinylcholine cause a transient increase in serum potassium, and in which patients is this response exaggerated and dangerous?

Answer

Succinylcholine depolarizes the entire postjunctional membrane and, in states of upregulated extrajunctional acetylcholine receptors (such as burns, denervation injuries, prolonged immobility, or neuromuscular disease), causes a massive efflux of potassium that can precipitate life-threatening hyperkalemia and cardiac arrest.

#35 · Basic Sciences

What is the mechanism of local anesthetic action on nerve conduction?

Answer

Local anesthetics reversibly bind the intracellular portion of voltage-gated sodium channels, blocking sodium influx and preventing the generation and propagation of action potentials along the nerve fiber.

#36 · Basic Sciences

What is the treatment of choice for local anesthetic systemic toxicity, and what is its proposed mechanism?

Answer

Intravenous lipid emulsion (20 percent) is the treatment of choice, working through a 'lipid sink' effect that sequesters lipophilic local anesthetic away from cardiac tissue, along with possible direct metabolic and cardiotonic effects.

#37 · Basic Sciences

How do genetic variants in butyrylcholinesterase (pseudocholinesterase) affect succinylcholine metabolism?

Answer

Patients homozygous for atypical pseudocholinesterase alleles metabolize succinylcholine very slowly, resulting in prolonged neuromuscular blockade that can last several hours instead of the usual few minutes.

#38 · Basic Sciences

On an arterial blood gas, how do you distinguish acute from chronic respiratory acidosis using the bicarbonate value?

Answer

In acute respiratory acidosis, renal compensation has not yet occurred, so bicarbonate rises only slightly (roughly 1 mEq/L per 10 mmHg rise in PaCO₂); in chronic respiratory acidosis, renal compensation raises bicarbonate substantially more (roughly 3 to 4 mEq/L per 10 mmHg rise).

#39 · Basic Sciences

What does the anion gap calculation tell you, and what is a common formula?

Answer

The anion gap, calculated as sodium minus the sum of chloride and bicarbonate, identifies unmeasured anions in the blood; an elevated gap suggests processes like lactic acidosis, ketoacidosis, uremia, or toxic ingestions.

#40 · Basic Sciences

On a capnography or blood gas analysis, what does a widened arterial to end-tidal CO₂ gradient suggest?

Answer

A widened PaCO₂ to ETCO₂ gradient suggests increased alveolar dead space, meaning ventilated alveoli are underperfused, as seen in pulmonary embolism, decreased cardiac output, or severe hypovolemia.

#41 · Basic Sciences

How should you interpret a sudden drop in SpO2 accompanied by a normal capnography waveform and stable end-tidal CO2?

Answer

A preserved normal capnography waveform with stable ETCO2 suggests adequate ventilation and circuit integrity, so the SpO2 drop is more likely due to a pulse oximeter probe or signal artifact rather than a true ventilation or perfusion catastrophe, though true hypoxemia from a V/Q mismatch cannot be excluded without further assessment.

#42 · Basic Sciences

On a processed EEG (such as bispectral index) monitor, what does a burst suppression pattern indicate, and in what clinical scenario is it intentionally targeted?

Answer

Burst suppression reflects a profoundly deep level of anesthesia or cerebral metabolic suppression, alternating high-amplitude bursts with isoelectric periods; it is intentionally targeted during therapeutic interventions such as barbiturate coma for refractory intracranial hypertension or status epilepticus.

#43 · Basic Sciences

How do you interpret a persistently elevated plateau pressure with a normal peak pressure change on a ventilator?

Answer

An isolated increase in plateau pressure with a proportionate change in peak pressure suggests decreased pulmonary or chest wall compliance, such as from pulmonary edema, ARDS, pneumothorax, or abdominal distension, rather than an airway resistance problem.

#44 · Basic Sciences

How do you interpret a rise in peak inspiratory pressure with an unchanged plateau pressure?

Answer

A rise in peak pressure with unchanged plateau pressure indicates increased airway resistance, such as bronchospasm, secretions, a kinked endotracheal tube, or a partially obstructed circuit, since resistive pressures affect peak but not plateau pressure.

#45 · Basic Sciences

On thromboelastography (TEG), what does a prolonged R time versus a decreased maximum amplitude (MA) each suggest, and how does treatment differ?

Answer

A prolonged R time reflects a deficiency of clotting factors and is treated with fresh frozen plasma, while a decreased maximum amplitude reflects poor platelet-fibrin clot strength, most often from platelet dysfunction or thrombocytopenia, and is treated with platelet transfusion.

#46 · Clinical Sciences

What is the mechanism of local anesthetic systemic toxicity (LAST) at the cellular level?

Answer

Local anesthetics block voltage-gated sodium channels in cardiac and CNS tissue, impairing conduction. At toxic plasma levels this produces CNS excitation followed by depression, and direct myocardial depression with conduction block leading to arrhythmias and cardiac arrest.

#47 · Clinical Sciences

What is the recommended initial bolus dose of 20% lipid emulsion for treating LAST in a patient over 70 kg?

Answer

1.5 mL/kg of 20% lipid emulsion given as an intravenous bolus over 2 to 3 minutes, followed by a continuous infusion, per ASRA LAST checklist guidance.

#48 · Clinical Sciences

Why should propofol not be substituted for lipid emulsion in the treatment of LAST?

Answer

Propofol contains far too little lipid to provide a therapeutic lipid effect, and its own cardiodepressant properties can worsen the hemodynamic instability already caused by local anesthetic toxicity.

#49 · Clinical Sciences

During LAST resuscitation, how should epinephrine dosing differ from standard ACLS dosing?

Answer

Epinephrine boluses should be reduced to less than 1 mcg/kg because larger doses can impair resuscitation with lipid emulsion and may worsen ventricular arrhythmias in the setting of local anesthetic-induced sodium channel blockade.

#50 · Clinical Sciences

Why is bupivacaine considered more cardiotoxic than lidocaine at equipotent local anesthetic doses?

Answer

Bupivacaine binds cardiac sodium channels with higher affinity and dissociates much more slowly than lidocaine, so it accumulates during diastole, producing a disproportionate risk of refractory ventricular arrhythmias relative to its CNS toxic dose.

#51 · Clinical Sciences

What early CNS signs suggest evolving local anesthetic systemic toxicity?

Answer

Perioral numbness, metallic taste, tinnitus, visual disturbance, and agitation or confusion typically precede overt seizure activity and should prompt immediate cessation of local anesthetic injection.

#52 · Clinical Sciences

According to ASRA guidelines, how many days should a direct oral anticoagulant like apixaban be held before neuraxial block in a patient with normal renal function?

Answer

Apixaban should generally be held for 3 days (72 hours) before neuraxial puncture or catheter removal, with a longer interval in patients with renal impairment.

#53 · Clinical Sciences

How long should therapeutic-dose low molecular weight heparin be held before neuraxial anesthesia?

Answer

At least 24 hours should elapse after the last therapeutic dose of low molecular weight heparin before neuraxial needle or catheter placement.

#54 · Clinical Sciences

What is the recommended interval between prophylactic-dose low molecular weight heparin and neuraxial catheter removal?

Answer

Neuraxial catheter removal should occur at least 12 hours after the last prophylactic dose of low molecular weight heparin, with the next dose delayed at least 4 hours after removal.

#55 · Clinical Sciences

How does unfractionated heparin management differ for subcutaneous prophylactic dosing versus intravenous therapeutic dosing before neuraxial block?

Answer

Subcutaneous prophylactic unfractionated heparin has no strict time-based contraindication per ASRA, though block after more than 4 days of therapy warrants checking platelet count, while intravenous therapeutic heparin should be held for 4 to 6 hours with a normal aPTT before neuraxial puncture.

#56 · Clinical Sciences

What is the ASRA recommendation regarding neuraxial anesthesia and clopidogrel?

Answer

Clopidogrel should be discontinued for 5 to 7 days before neuraxial block placement to allow adequate platelet function recovery, given its irreversible P2Y₁₂ receptor inhibition.

#57 · Clinical Sciences

What clinical presentation should raise suspicion for spinal epidural hematoma after neuraxial anesthesia?

Answer

New or progressive back pain, sensory or motor deficit exceeding what is expected from the local anesthetic, and bowel or bladder dysfunction; this is a surgical emergency requiring urgent MRI and neurosurgical decompression within 8 hours for the best chance of neurologic recovery.

#58 · Clinical Sciences

What distinguishes the mechanism of a spinal (subarachnoid) block from an epidural block?

Answer

A spinal block deposits local anesthetic directly into the cerebrospinal fluid in the subarachnoid space, producing dense rapid-onset blockade with a small dose, whereas an epidural block deposits local anesthetic into the epidural space where it must diffuse across the dura, producing a slower onset segmental block that is more easily titrated and can be extended through a catheter.

#59 · Clinical Sciences

Why does spinal anesthesia typically cause more pronounced hypotension than a comparable epidural block?

Answer

Spinal anesthesia produces rapid, dense sympathetic blockade that extends 2 or more dermatomal levels above the sensory block, causing abrupt venodilation and decreased venous return, whereas epidural onset is more gradual, allowing better compensatory hemodynamic adaptation.

#60 · Clinical Sciences

What is the mechanism of post-dural puncture headache and why is it worse when upright?

Answer

Dural puncture allows continued cerebrospinal fluid leak, lowering intracranial CSF pressure and causing traction on pain-sensitive meninges and compensatory cerebral vasodilation; sitting or standing increases CSF hydrostatic loss and worsens the traction, so the headache classically improves when supine.

#61 · Clinical Sciences

What is the definitive treatment for a severe or persistent post-dural puncture headache?

Answer

An epidural blood patch, in which 15 to 20 mL of the patient's own autologous blood is injected into the epidural space near the puncture site, sealing the dural defect and restoring CSF pressure, providing relief in the majority of patients.

#62 · Clinical Sciences

What needle characteristic most reduces the incidence of post-dural puncture headache?

Answer

Using a smaller gauge, pencil-point (atraumatic) spinal needle such as a Whitacre or Sprotte tip, which spreads dural fibers rather than cutting them, substantially lowers PDPH incidence compared with cutting-tip Quincke needles.

#63 · Clinical Sciences

What defines high spinal or total spinal anesthesia and what are its cardinal signs?

Answer

Total spinal anesthesia results from excessive cephalad spread of local anesthetic in the subarachnoid space, blocking cervical and even brainstem levels, producing bradycardia, severe hypotension, apnea from phrenic and intercostal paralysis, and loss of consciousness requiring immediate airway support and hemodynamic resuscitation.

#64 · Clinical Sciences

How does baricity of a local anesthetic solution influence spread during spinal anesthesia?

Answer

Hyperbaric solutions (mixed with dextrose) are denser than CSF and settle with gravity to dependent areas, hypobaric solutions are less dense and rise, and isobaric solutions remain near the level of injection, allowing patient positioning to control the level of block with hyperbaric or hypobaric agents.

#65 · Clinical Sciences

What is transient neurologic symptoms (TNS) and which local anesthetic is most associated with it?

Answer

TNS presents as bilateral buttock, thigh, or lower back pain appearing within 24 hours of spinal anesthesia and resolving spontaneously within a week; it is most strongly associated with spinal lidocaine, particularly in the lithotomy position.

#66 · Clinical Sciences

What are the key sonographic landmarks used to identify the interscalene brachial plexus block target?

Answer

The brachial plexus roots and trunks appear as a stack of round hypoechoic structures ('traffic light sign') between the anterior and middle scalene muscles at the level of the cricoid cartilage, typically superficial to the lateral border of the sternocleidomastoid.

#67 · Clinical Sciences

What complication is nearly universal with an interscalene brachial plexus block and why?

Answer

Ipsilateral hemidiaphragmatic paresis from phrenic nerve blockade occurs in nearly all patients because the phrenic nerve runs on the anterior surface of the anterior scalene muscle immediately adjacent to the injection site, which can be significant in patients with limited pulmonary reserve.

#68 · Clinical Sciences

Which peripheral nerve block is preferred for anesthesia or analgesia of the distal forearm and hand while sparing the phrenic nerve?

Answer

A supraclavicular, infraclavicular, or axillary brachial plexus block is preferred over an interscalene block for distal upper extremity surgery, since these lower approaches carry a much lower risk of phrenic nerve involvement.

#69 · Clinical Sciences

What structure defines the supraclavicular brachial plexus block target and what serious risk is uniquely associated with this approach?

Answer

The brachial plexus trunks and divisions are compactly grouped lateral and superior to the subclavian artery at the level of the first rib ('corner pocket'); the close proximity of the pleura at this level carries a risk of pneumothorax.

#70 · Clinical Sciences

Which nerve block provides analgesia to the anterior hip capsule and femoral shaft while largely sparing motor function, useful for hip fracture pain?

Answer

The fascia iliaca compartment block, which spreads local anesthetic deep to the fascia iliaca to bathe the femoral, lateral femoral cutaneous, and obturator nerves, providing analgesia with relatively preserved quadriceps strength compared with a femoral nerve block.

#71 · Clinical Sciences

Why has the adductor canal block largely replaced the femoral nerve block for postoperative analgesia after total knee arthroplasty?

Answer

The adductor canal block targets the predominantly sensory saphenous nerve and articular branches distal to the femoral triangle, providing comparable analgesia to a femoral block while preserving quadriceps motor strength, which facilitates earlier ambulation and reduces fall risk.

#72 · Clinical Sciences

What is the sonoanatomy target for a popliteal sciatic nerve block?

Answer

The sciatic nerve is identified in the popliteal fossa proximal to its bifurcation into the tibial and common peroneal nerves, typically visualized posterior to the popliteal artery and vein, superficial to the biceps femoris and semimembranosus muscles.

#73 · Clinical Sciences

What is the mechanism and clinical utility of the transversus abdominis plane (TAP) block?

Answer

Local anesthetic is deposited in the fascial plane between the internal oblique and transversus abdominis muscles, blocking the thoracolumbar sensory nerves (T7 through L1) that travel in this plane, providing somatic analgesia to the anterior abdominal wall for procedures such as cesarean delivery and abdominal surgery, though it does not cover visceral pain.

#74 · Clinical Sciences

How does the erector spinae plane (ESP) block achieve analgesic spread and what is its proposed mechanism?

Answer

Local anesthetic is injected deep to the erector spinae muscle onto the transverse process, spreading craniocaudally in the fascial plane to reach the dorsal and ventral rami of spinal nerves, and possibly to the paravertebral space, providing multidermatomal thoracic or abdominal analgesia.

#75 · Clinical Sciences

What is the primary safety advantage of real-time ultrasound guidance over a nerve stimulator technique for peripheral nerve blockade?

Answer

Ultrasound allows direct visualization of the needle tip, target nerve, and surrounding vascular structures during advancement and injection, reducing the risk of intraneural or intravascular injection and improving block success while potentially lowering the volume of local anesthetic required.

#76 · Clinical Sciences

What ultrasound finding during peripheral nerve block injection should prompt immediate needle repositioning?

Answer

Failure to visualize local anesthetic spreading around the nerve, or a marked increase in nerve cross-sectional area with hypoechoic swelling during injection, suggests intraneural injection and mandates stopping injection and repositioning the needle to avoid nerve injury.

#77 · Clinical Sciences

What defines total intravenous anesthesia (TIVA) and what are its two most commonly combined agents?

Answer

TIVA is general anesthesia maintained entirely with intravenous agents without inhaled volatile anesthetic, most commonly combining a continuous propofol infusion for hypnosis with a short-acting opioid such as remifentanyl for analgesia, often titrated using target-controlled infusion or processed EEG monitoring.

#78 · Clinical Sciences

What is a key clinical advantage of TIVA in patients with a history of postoperative nausea and vomiting?

Answer

Propofol has intrinsic antiemetic properties and avoiding volatile anesthetics eliminates a major contributor to PONV, making TIVA a preferred technique for patients at high risk for postoperative nausea and vomiting.

#79 · Clinical Sciences

Why is remifentanyl particularly well suited to TIVA regimens?

Answer

Remifentanyl is metabolized by nonspecific plasma and tissue esterases independent of hepatic or renal function, giving it an extremely short and context-insensitive half-life that allows rapid titration and predictable, quick offset even after prolonged infusion.

#80 · Clinical Sciences

What is the definition of monitored anesthesia care (MAC) and how does it differ from moderate sedation?

Answer

MAC is a specific anesthesia service in which an anesthesia provider monitors the patient and is prepared to convert to general or regional anesthesia or manage any level of sedation as needed, including deep sedation or general anesthesia if required, distinguishing it from moderate sedation which does not require the same level of readiness or vigilance.

#81 · Clinical Sciences

What perioperative goals must be balanced during an awake craniotomy for tumor resection near eloquent cortex?

Answer

The patient must be adequately sedated and comfortable with a well-anesthetized scalp during opening and closing, yet fully awake, cooperative, and neurologically testable during cortical mapping and resection, typically achieved with an asleep-awake-asleep technique or conscious sedation combined with a scalp block.

#82 · Clinical Sciences

Why is a scalp block particularly valuable in awake craniotomy technique?

Answer

A scalp block anesthetizes the sensory nerves supplying the scalp (supraorbital, supratrochlear, zygomaticotemporal, auriculotemporal, greater and lesser occipital nerves), providing excellent analgesia for pin placement, incision, and closure while minimizing the systemic sedative or opioid requirement that could impair intraoperative neurologic testing.

#83 · Clinical Sciences

What is the physiologic basis for hypoxic pulmonary vasoconstriction and why does it matter during one-lung ventilation?

Answer

Hypoxic pulmonary vasoconstriction diverts pulmonary blood flow away from poorly ventilated alveoli toward better-ventilated regions, and during one-lung ventilation this reflex redirects blood flow from the collapsed non-dependent lung to the ventilated dependent lung, helping to limit shunt and preserve oxygenation.

#84 · Clinical Sciences

How do volatile anesthetics affect hypoxic pulmonary vasoconstriction during one-lung ventilation?

Answer

Volatile anesthetics blunt hypoxic pulmonary vasoconstriction in a dose-dependent manner, which can worsen shunt and oxygenation during one-lung ventilation, favoring the use of lower volatile concentrations or a TIVA-based technique when oxygenation is marginal.

#85 · Clinical Sciences

What are the first steps in managing hypoxemia during one-lung ventilation?

Answer

Confirm correct double-lumen tube or bronchial blocker position with fiberoptic bronchoscopy, increase the FiO₂ to 100%, apply CPAP to the non-dependent (non-ventilated) lung, and apply PEEP to the dependent lung, escalating to intermittent two-lung ventilation if hypoxemia persists.

#86 · Clinical Sciences

What is the physiologic rationale for using deliberate (controlled) hypotension during select surgical procedures?

Answer

Deliberate hypotension reduces mean arterial pressure to decrease surgical field bleeding and improve visualization, most commonly used in procedures such as spine surgery, functional endoscopic sinus surgery, and some orthopedic and plastic procedures, while requiring careful attention to maintain adequate perfusion to vital organs.

#87 · Clinical Sciences

What is the generally accepted lower limit for mean arterial pressure during deliberate hypotensive anesthesia in a healthy patient, and why must this be individualized?

Answer

A mean arterial pressure of roughly 50 to 65 mmHg is often targeted, but the true safe limit depends on the patient's baseline blood pressure, cerebral and coronary autoregulation, and comorbidities, since patients with chronic hypertension have a rightward-shifted autoregulatory curve and tolerate hypotension poorly.

#88 · Clinical Sciences

What is the role of the oxygenator in cardiopulmonary bypass (CPB)?

Answer

The oxygenator replaces pulmonary gas exchange during bypass by adding oxygen to and removing carbon dioxide from venous blood before it is returned to the arterial circulation, allowing the heart and lungs to be excluded from the circulation during cardiac surgery.

#89 · Clinical Sciences

Why is systemic heparinization to an activated clotting time (ACT) target required before initiating cardiopulmonary bypass?

Answer

Blood contact with the large non-endothelialized surface area of the bypass circuit is intensely thrombogenic, so full anticoagulation, typically confirmed by an ACT greater than 400 to 480 seconds, is required before cannulation to prevent circuit thrombosis and consumptive coagulopathy.

#90 · Clinical Sciences

How is heparin's anticoagulant effect reversed after separation from cardiopulmonary bypass, and what determines the reversal dose?

Answer

Protamine sulfate is administered to neutralize heparin, typically dosed based on the estimated residual circulating heparin dose (often using a 1:1 mg ratio to the total heparin given, adjusted by protocol or point-of-care testing), with the ACT rechecked afterward to confirm adequate reversal.

#91 · Clinical Sciences

What hemodynamic reaction can occur with rapid protamine administration and what is the underlying mechanism?

Answer

Rapid protamine administration can cause severe systemic hypotension from direct histamine release and vasodilation, and in some patients can trigger catastrophic pulmonary vasoconstriction and right heart failure from complement-mediated thromboxane release, so protamine should be given slowly with hemodynamic monitoring.

#92 · Clinical Sciences

According to ASA difficult airway algorithm principles, what should be done if mask ventilation fails after induction and intubation attempts are unsuccessful?

Answer

Attempt placement of a supraglottic airway device as a rescue technique to restore ventilation and oxygenation; if this also fails and ventilation cannot be achieved, this constitutes a 'cannot intubate, cannot oxygenate' emergency requiring immediate surgical or percutaneous airway access.

#93 · Clinical Sciences

What is the definitive emergency airway rescue technique when a 'cannot intubate, cannot oxygenate' scenario is declared?

Answer

An emergency invasive airway, most commonly a surgical cricothyrotomy, is performed to establish an airway below the level of obstruction, and this decision should be made early rather than after repeated failed attempts at less invasive rescue techniques.

#94 · Clinical Sciences

What is the underlying genetic and cellular defect responsible for malignant hyperthermia?

Answer

Malignant hyperthermia results from a mutation, most commonly in the RYR1 gene encoding the ryanodine receptor, causing uncontrolled release of calcium from the sarcoplasmic reticulum of skeletal muscle when triggered by volatile anesthetics or succinylcholine, leading to sustained muscle contraction, hypermetabolism, and rhabdomyolysis.

#95 · Clinical Sciences

What is the immediate first-line pharmacologic treatment for an acute malignant hyperthermia crisis and how does it work?

Answer

Dantrolene sodium is given immediately, starting at 2.5 mg/kg IV and repeated as needed, because it inhibits calcium release from the ryanodine receptor in the sarcoplasmic reticulum, reducing myoplasmic calcium and reversing the hypermetabolic muscle contracture.

#96 · Organ-Based Sciences

What is the normal range of mean arterial pressure over which cerebral autoregulation maintains constant cerebral blood flow?

Answer

Roughly 60 to 150 mmHg MAP in a normotensive patient. Below the lower limit CBF falls passively with pressure (risk of ischemia), and above the upper limit CBF rises passively (risk of edema and hemorrhage).

#97 · Organ-Based Sciences

How is cerebral perfusion pressure calculated, and what value is generally targeted after traumatic brain injury?

Answer

CPP equals mean arterial pressure minus intracranial pressure (or central venous pressure if higher). A CPP of about 60 to 70 mmHg is generally targeted to balance adequate perfusion against worsening cerebral edema.

#98 · Organ-Based Sciences

How do volatile anesthetics affect the normal coupling between cerebral metabolic rate and cerebral blood flow?

Answer

Volatile agents cause dose-dependent cerebral vasodilation while simultaneously decreasing cerebral metabolic rate for oxygen, uncoupling flow from metabolism. At higher concentrations the vasodilating effect predominates, increasing cerebral blood flow and potentially intracranial pressure despite reduced metabolic demand.

#99 · Organ-Based Sciences

Describe the cerebral vasculature's response to changes in arterial CO₂, and over what PaCO₂ range is this response linear.

Answer

Cerebral vessels dilate with hypercapnia and constrict with hypocapnia, and CBF changes roughly 1 to 2 mL per 100 g per minute for each mmHg change in PaCO₂. This CO₂ reactivity is essentially linear between PaCO₂ of about 20 and 80 mmHg.

#100 · Organ-Based Sciences

By what mechanism does hyperventilation reduce intracranial pressure, and why is its benefit time-limited?

Answer

Hyperventilation lowers PaCO₂, causing cerebral vasoconstriction and reduced cerebral blood volume, which lowers ICP within minutes. The effect wanes over 6 to 24 hours as cerebrospinal fluid bicarbonate re-equilibrates and vascular tone normalizes, so prolonged aggressive hyperventilation can also risk cerebral ischemia.

#101 · Organ-Based Sciences

What is the Cushing triad and what does it indicate?

Answer

The Cushing triad is hypertension, bradycardia, and irregular respirations. It reflects the body's compensatory response to critically elevated intracranial pressure and impending brainstem herniation.

#102 · Organ-Based Sciences

What are the mechanisms by which mannitol reduces intracranial pressure?

Answer

Mannitol acutely expands plasma volume and reduces blood viscosity, improving cerebral blood flow and triggering reflex vasoconstriction, then acts as an osmotic diuretic that draws water out of brain tissue across an intact blood-brain barrier over 15 to 30 minutes.

#103 • Organ-Based Sciences

What features make an induction agent favorable for a patient with elevated intracranial pressure?

Answer

An ideal agent reduces cerebral metabolic rate and cerebral blood volume (lowering ICP) while maintaining cerebral perfusion pressure and hemodynamic stability. Propofol and etomidate are commonly favored, while ketamine (without co-administered sedative-hypnotics) is traditionally avoided due to cerebral vasodilation.

#104 • Organ-Based Sciences

Does succinylcholine cause a clinically significant rise in intracranial pressure?

Answer

Succinylcholine may cause a small, transient increase in ICP, likely related to increased muscle afferent activity and light anesthesia rather than a direct effect, but current evidence does not support withholding it when rapid, reliable intubation is needed in a patient with elevated ICP.

#105 • Organ-Based Sciences

Why is nitrous oxide generally avoided during craniotomy for tumor resection or after recent intracranial surgery?

Answer

Nitrous oxide is a cerebral vasodilator that can increase cerebral blood flow and intracranial pressure, and it can expand any trapped intracranial air (pneumocephalus), worsening mass effect.

#106 • Organ-Based Sciences

What is the most sensitive intraoperative monitor for detecting venous air embolism during a sitting craniotomy?

Answer

Precordial Doppler ultrasound is the most sensitive routinely used monitor, detecting air volumes as small as 0.25 mL, more sensitive than end-tidal CO₂ decrease, transesophageal echocardiography being even more sensitive but less practical for continuous monitoring.

#107 • Organ-Based Sciences

What immediate steps should be taken when venous air embolism is suspected during neurosurgery?

Answer

Notify the surgeon to flood the field and apply bone wax or occlude the entry site, discontinue nitrous oxide, give 100% oxygen, aspirate air through a previously placed multi-orifice central venous catheter, and support blood pressure with fluids and vasopressors; left lateral (Durant) positioning has unproven benefit and is not routinely required.

#108 • Organ-Based Sciences

What are the major physiologic risks specific to the sitting position for craniotomy?

Answer

The sitting position increases the risk of venous air embolism due to a negative pressure gradient between the surgical site and the heart, and it predisposes to significant hypotension from venous pooling, along with risks of macroglossia and peripheral nerve injury.

#109 • Organ-Based Sciences

What triggers autonomic dysreflexia in a patient with chronic spinal cord injury, and at what injury level does it typically occur?

Answer

Autonomic dysreflexia occurs with spinal cord lesions at or above T6, triggered by noxious stimuli below the level of injury (commonly bladder distension or bowel manipulation), causing unopposed sympathetic outflow below the lesion with severe hypertension and reflex bradycardia above it.

#110 • Organ-Based Sciences

Distinguish neurogenic shock from spinal shock after acute spinal cord injury.

Answer

Neurogenic shock refers to hemodynamic instability (hypotension and bradycardia) from loss of sympathetic tone below the injury level, while spinal shock refers to the transient loss of all reflexes, sensation, and motor function below the lesion regardless of hemodynamics; the two often coexist but describe different phenomena.

#111 • Organ-Based Sciences

Why is succinylcholine avoided in patients with spinal cord injury, and during what time window does the risk apply?

Answer

Denervation causes proliferation of extrajunctional acetylcholine receptors, and succinylcholine administration can trigger exaggerated potassium efflux and life-threatening hyperkalemia. The risk begins around 24 to 48 hours after injury and can persist for many months to years.

#112 • Organ-Based Sciences

What blood pressure strategy is generally recommended during the acute phase of an ischemic stroke?

Answer

Permissive hypertension is generally allowed to maintain perfusion to the ischemic penumbra through collateral circulation, since aggressive blood pressure lowering can extend the area of infarction; specific thresholds depend on whether thrombolysis or thrombectomy is performed.

#113 • Organ-Based Sciences

What perioperative measures are supported as neuroprotective strategies during periods of cerebral ischemia risk?

Answer

Maintaining normothermia (avoiding hyperthermia), tight glucose control avoiding both hyperglycemia and hypoglycemia, and maintaining adequate cerebral perfusion pressure are supported strategies; induced moderate hypothermia and barbiturate coma are reserved for specific refractory situations with less robust supporting evidence.

#114 • Organ-Based Sciences

What is the proposed mechanism by which high-dose barbiturates provide cerebral protection?

Answer

Barbiturates reduce cerebral metabolic rate for oxygen and cerebral blood flow in a dose-dependent manner up to the point of EEG burst suppression, theoretically reducing energy demand during periods of inadequate substrate delivery.

#115 • Organ-Based Sciences

What does EEG burst suppression represent, and what is it used to guide?

Answer

Burst suppression is an EEG pattern of alternating bursts of electrical activity and periods of electrical silence, reflecting maximal pharmacologic suppression of cerebral metabolic activity. It is used to titrate anesthetic or barbiturate infusions to their maximal metabolic-suppressing effect.

#116 • Organ-Based Sciences

Which volatile anesthetic has historically been associated with epileptiform EEG activity at high concentrations, particularly with hypocapnia?

Answer

Sevoflurane can produce epileptiform spikes on EEG at high inspired concentrations, especially combined with hyperventilation-induced hypocapnia; enflurane, now rarely used, was also well known for this effect.

#117 · Organ-Based Sciences

What EEG and clinical phenomenon is commonly seen with etomidate induction?

Answer

Etomidate frequently causes myoclonic movements and can produce epileptiform activity on EEG, though it does not reliably lower the seizure threshold clinically and has even been used to help localize epileptogenic foci during electrocorticography.

#118 · Organ-Based Sciences

What is the general anesthetic approach for an awake craniotomy with intraoperative cortical mapping?

Answer

An asleep-awake-asleep or monitored anesthesia care technique is used, with scalp block and local infiltration for pain control, sedation with short-acting agents such as propofol and remifentanyl for the asleep phases, and full patient cooperation during the awake mapping phase, avoiding drugs that suppress cortical function during testing.

#119 · Organ-Based Sciences

How do volatile anesthetics generally affect somatosensory and motor evoked potentials used for intraoperative neuromonitoring?

Answer

Volatile anesthetics cause dose-dependent decreases in amplitude and increases in latency of both SSEPs and MEPs, with motor evoked potentials being markedly more sensitive to this suppression; total intravenous anesthesia with propofol and opioid is generally preferred when motor monitoring is critical.

#120 · Organ-Based Sciences

What mean arterial pressure goal is typically recommended in the days following acute traumatic spinal cord injury, and why?

Answer

A MAP goal of at least 85 to 90 mmHg is typically maintained for several days to optimize spinal cord perfusion pressure and limit secondary ischemic injury to the cord.

#121 · Organ-Based Sciences

Describe the classic West lung zone model of ventilation-perfusion matching.

Answer

In the upright lung, zone 1 (apex) has ventilation exceeding perfusion due to low pulmonary arterial pressure, zone 2 (mid-lung) has better matching, and zone 3 (base) has perfusion exceeding ventilation due to gravity-dependent increased blood flow, though ventilation also increases toward the base.

#122 · Organ-Based Sciences

What is hypoxic pulmonary vasoconstriction and what agents inhibit it?

Answer

Hypoxic pulmonary vasoconstriction is a local response in which alveolar hypoxia causes constriction of the adjacent pulmonary arteriole, diverting blood flow away from poorly ventilated lung units to improve overall V/Q matching. Volatile anesthetics (at higher concentrations), vasodilators, and calcium channel blockers all blunt this protective reflex.

#123 · Organ-Based Sciences

How do obstructive and restrictive patterns differ on spirometry, particularly the FEV1/FVC ratio?

Answer

Obstructive disease shows a reduced FEV1/FVC ratio (typically below 0.7) with a normal or increased total lung capacity, reflecting airflow limitation. Restrictive disease shows a preserved or increased FEV1/FVC ratio with reduced FVC and total lung capacity, reflecting reduced lung volume.

#124 · Organ-Based Sciences

How does the flow-volume loop differ between obstructive and restrictive lung disease?

Answer

Obstructive disease shows a scooped-out, concave expiratory limb with reduced peak expiratory flow and prolonged expiration. Restrictive disease shows a normal or steep shape but with an overall narrowed loop reflecting reduced lung volumes.

#125 • Organ-Based Sciences

What is the underlying pathophysiology of bronchospasm in an acute asthma exacerbation?

Answer

Airway smooth muscle contraction, mucosal edema, and increased mucus secretion combine to narrow the airway lumen, driven by an inflammatory cascade involving mast cell degranulation, histamine and leukotriene release, and bronchial hyperreactivity to triggers.

#126 • Organ-Based Sciences

What anesthetic drug choices should generally be avoided in a patient with reactive airway disease, and why?

Answer

Histamine-releasing drugs such as morphine and atracurium, and drugs associated with airway irritation such as desflurane during induction or light-plane instrumentation, should generally be avoided since they can precipitate bronchospasm; propofol and volatile agents like sevoflurane are preferred for their bronchodilating properties.

#127 · Organ-Based Sciences

What perioperative factors are most predictive of postoperative pulmonary complications in a patient with COPD?

Answer

Poor functional status, low preoperative FEV1, active smoking, upper abdominal or thoracic surgical site, prolonged anesthesia duration, and general anesthesia (versus regional when feasible) are among the strongest predictors of postoperative pulmonary complications.

#128 · Organ-Based Sciences

What is the definition of obstructive sleep apnea and what does the STOP-BANG questionnaire assess?

Answer

OSA is characterized by repetitive partial or complete upper airway obstruction during sleep causing apneas or hypopneas with oxygen desaturation. STOP-BANG assesses Snoring, Tiredness, Observed apnea, high blood Pressure, BMI over 35, Age over 50, Neck circumference over 40 cm, and male Gender, with higher scores indicating greater risk of moderate to severe OSA.

#129 · Organ-Based Sciences

What are the key perioperative implications of obstructive sleep apnea?

Answer

Patients with OSA have increased sensitivity to sedatives and opioids with higher risk of airway obstruction and respiratory depression, increased likelihood of difficult mask ventilation and intubation, and higher risk of postoperative hypoxemic events, favoring opioid-sparing multimodal analgesia and extended postoperative monitoring.

#130 · Organ-Based Sciences

How is sleep apnea severity classified using the apnea-hypopnea index?

Answer

An AHI of 5 to 14 events per hour is mild, 15 to 29 is moderate, and 30 or more events per hour is classified as severe obstructive sleep apnea.

#131 • Organ-Based Sciences

What are the primary indications for one-lung ventilation?

Answer

Absolute indications include isolating a source of infection or bleeding to protect the healthy lung, controlling distribution of ventilation such as with a bronchopleural fistula or giant bullae, and unilateral lung lavage. Relative indications include facilitating surgical exposure for thoracic, esophageal, or thoracoscopic procedures.

#132 • Organ-Based Sciences

During one-lung ventilation in the lateral decubitus position under general anesthesia with an open chest, how is blood flow distributed between the lungs and why?

Answer

The dependent lung receives greater perfusion due to gravity, but under general anesthesia with paralysis and an open nondependent hemithorax, the dependent lung also bears most of the ventilatory work and compliance burden, worsening V/Q mismatch compared to the awake, spontaneously breathing state.

#133 · Organ-Based Sciences

What stepwise measures are used to treat hypoxemia during one-lung ventilation?

Answer

Increase FiO₂ to 100%, confirm correct double-lumen tube or blocker position with fiberoptic bronchoscopy, apply CPAP to the nondependent (collapsed) lung, apply PEEP to the dependent ventilated lung, and if hypoxemia persists, intermittently reinflate the nondependent lung or ask the surgeon to briefly clamp the pulmonary artery to the operative lung.

#134 · Organ-Based Sciences

How is correct double-lumen tube position confirmed intraoperatively?

Answer

Fiberoptic bronchoscopy through both the tracheal and bronchial lumens is the gold standard, confirming the bronchial cuff sits just below the carina in the correct mainstem bronchus and that the tracheal opening lies above the carina; auscultation alone is unreliable for confirming position.

#135 · Organ-Based Sciences

What are the Berlin criteria used to define acute respiratory distress syndrome?

Answer

ARDS requires acute onset within one week of a known insult, bilateral opacities on chest imaging not fully explained by effusion, lobar collapse, or nodules, respiratory failure not fully explained by cardiac failure or fluid overload, and a $\text{PaO}_2/\text{FiO}_2$ ratio of 300 or less with at least 5 cmH₂O of PEEP, with severity graded as mild, moderate, or severe based on that ratio.

#136 · Organ-Based Sciences

What are the core components of lung-protective ventilation in ARDS?

Answer

Low tidal volume of about 6 mL/kg predicted body weight, limiting plateau pressure to 30 cmH₂O or less, and using adequate PEEP to prevent alveolar derecruitment, all aimed at minimizing ventilator-induced lung injury from overdistension and cyclic atelectasis.

#137 • Organ-Based Sciences

What is permissive hypercapnia and when is it accepted during mechanical ventilation for ARDS?

Answer

Permissive hypercapnia is the deliberate tolerance of elevated PaCO₂ and resulting mild respiratory acidosis that results from using low tidal volumes to limit lung injury; it is generally accepted as a favorable tradeoff in ARDS unless contraindicated by conditions like elevated intracranial pressure or severe pulmonary hypertension.

#138 • Organ-Based Sciences

By what mechanism does PEEP improve oxygenation in ARDS?

Answer

PEEP increases functional residual capacity and recruits and stabilizes collapsed or unstable alveoli, reducing cyclic opening and closing (atelectrauma) and improving overall ventilation-perfusion matching.

#139 · Organ-Based Sciences

What is the physiologic rationale for prone positioning in severe ARDS?

Answer

Prone positioning improves ventilation-perfusion matching by more evenly distributing pleural pressure and lung aeration, recruiting dorsal lung regions that are typically dependent and atelectatic in the supine position, which improves oxygenation and has been shown to reduce mortality in severe ARDS.

#140 · Organ-Based Sciences

Give examples of restrictive lung disease and describe their underlying pathophysiology.

Answer

Restrictive diseases include pulmonary fibrosis, sarcoidosis, and chest wall or neuromuscular disorders such as kyphoscoliosis or myasthenia gravis. Parenchymal restrictive disease reduces lung compliance from fibrosis or infiltration, while extrapulmonary causes limit chest wall or diaphragmatic excursion, both reducing lung volumes.

#141 • Organ-Based Sciences

How does the relationship between closing capacity and functional residual capacity change with age and obesity, and why does this matter perioperatively?

Answer

Closing capacity increases with age while FRC tends to decrease with obesity and supine positioning; when closing capacity exceeds FRC, small airways close during normal tidal breathing, promoting atelectasis and increasing the risk of perioperative hypoxemia.

#142 • Organ-Based Sciences

What is the initial anesthetic management of intraoperative bronchospasm?

Answer

Deepen anesthesia with a volatile agent, ensure 100% oxygen, rule out mechanical causes such as a kinked or malpositioned endotracheal tube, and administer a beta-2 agonist such as albuterol via the circuit; intravenous epinephrine is reserved for severe, refractory bronchospasm.

#143 • Organ-Based Sciences

How does diffusing capacity for carbon monoxide (DLCO) typically differ between obstructive and restrictive lung disease?

Answer

DLCO is characteristically reduced in emphysema due to alveolar-capillary destruction but may be normal or increased in chronic bronchitis or asthma. In restrictive disease, DLCO is reduced in parenchymal causes like pulmonary fibrosis but tends to be normal in purely extrapulmonary restrictive disease such as chest wall deformity.

#144 • Organ-Based Sciences

What is auto-PEEP (breath stacking) and how is it managed in a mechanically ventilated patient with severe bronchospasm?

Answer

Auto-PEEP occurs when incomplete exhalation causes progressive air trapping and rising end-expiratory alveolar pressure, risking barotrauma and hemodynamic compromise from reduced venous return. Management includes reducing respiratory rate, prolonging expiratory time, treating bronchospasm, and briefly disconnecting the circuit to allow full exhalation if severe hypotension develops.

#145 • Organ-Based Sciences

Which preoperative factors are most strongly associated with increased pulmonary risk across surgical populations?

Answer

Advanced age, poor functional capacity, ASA class 3 or higher, heart failure, chronic obstructive pulmonary disease, and the planned surgical site (thoracic and upper abdominal procedures carry the highest risk) are among the strongest predictors of postoperative pulmonary complications.

#146 • Organ-Based Sciences

What are the six risk factors in the Revised Cardiac Risk Index (RCRI)?

Answer

High-risk surgery, history of ischemic heart disease, history of congestive heart failure, history of cerebrovascular disease, insulin-treated diabetes mellitus, and preoperative creatinine greater than 2.0 mg/dL. Each factor scores one point toward predicting perioperative cardiac risk.

#147 · Organ-Based Sciences

What functional capacity threshold, expressed in METs, is used to determine the need for further preoperative cardiac testing?

Answer

A functional capacity of at least 4 METs, such as climbing two flights of stairs or walking briskly on level ground, generally indicates adequate reserve and allows most patients to proceed to surgery without further cardiac workup.

#148 · Organ-Based Sciences

How long should elective noncardiac surgery be delayed after placement of a bare metal coronary stent?

Answer

At least 30 days, to allow completion of the minimum course of dual antiplatelet therapy and stent endothelialization before discontinuing an antiplatelet agent.

#149 • Organ-Based Sciences

How long should elective noncardiac surgery ideally be delayed after placement of a drug-eluting coronary stent?

Answer

Ideally at least 6 months, though surgery may proceed after 3 months if the risk of delaying surgery outweighs the risk of stent thrombosis, per a shared decision between surgeon, cardiologist, and anesthesiologist.

#150 • Organ-Based Sciences

What is the standard perioperative management of dual antiplatelet therapy in a patient with a recent coronary stent needing urgent surgery?

Answer

Aspirin is typically continued through the perioperative period, while the P2Y12 inhibitor (such as clopidogrel or ticagrelor) is held for 5 to 7 days preoperatively if surgical bleeding risk is high, ideally after consultation with cardiology given the stent thrombosis risk.

#151 · Organ-Based Sciences

What did the POISE trial demonstrate about initiating perioperative beta blockade?

Answer

High-dose metoprolol started shortly before noncardiac surgery reduced perioperative myocardial infarction but increased overall mortality and stroke, largely due to bradycardia and hypotension; this argued against routine initiation of beta blockers immediately before surgery.

#152 · Organ-Based Sciences

What is the current recommendation regarding chronic beta blocker therapy on the day of noncardiac surgery?

Answer

Beta blockers that a patient is already taking chronically should be continued perioperatively, but new beta blocker therapy should not be started on the day of surgery.

#153 • Organ-Based Sciences

What did the CARP trial show about prophylactic coronary revascularization before major vascular surgery?

Answer

Prophylactic coronary artery revascularization (PCI or CABG) before elective vascular surgery did not improve long-term survival or reduce perioperative myocardial infarction compared with medical therapy alone, except in patients with left main disease or severe unstable symptoms.

#154 • Organ-Based Sciences

When is preoperative noninvasive cardiac stress testing indicated before noncardiac surgery?

Answer

Testing is reserved for patients with elevated perioperative risk, poor or unknown functional capacity, and for whom the result would change perioperative management, rather than as routine screening for all at-risk patients.

#155 • Organ-Based Sciences

How is heart failure with reduced ejection fraction (HFrEF) distinguished from heart failure with preserved ejection fraction (HFpEF)?

Answer

HFrEF is defined by a left ventricular ejection fraction of 40% or less with impaired systolic contraction, while HFpEF has a normal or near-normal ejection fraction (generally 50% or greater) with impaired diastolic relaxation and filling.

#156 • Organ-Based Sciences

How is preoperative BNP or NT-proBNP used in cardiac risk stratification?

Answer

An elevated preoperative BNP or NT-proBNP identifies patients at increased risk of perioperative myocardial infarction, heart failure, and death after noncardiac surgery, and can be combined with the RCRI to refine risk estimates and guide postoperative troponin surveillance.

#157 · Organ-Based Sciences

What are the anesthetic hemodynamic goals for a patient with severe aortic stenosis?

Answer

Maintain sinus rhythm, adequate preload, normal to slightly elevated systemic vascular resistance, and avoid tachycardia and hypotension, since the fixed obstruction limits the ability to increase cardiac output and coronary perfusion depends on diastolic pressure gradient.

#158 · Organ-Based Sciences

What are the anesthetic hemodynamic goals for a patient with chronic aortic regurgitation?

Answer

Maintain a relatively fast heart rate (to shorten diastole and limit regurgitant flow), avoid increases in systemic vascular resistance, and maintain adequate preload and contractility, favoring a full, fast, and forward circulation.

#159 · Organ-Based Sciences

What are the anesthetic hemodynamic goals for a patient with mitral stenosis?

Answer

Maintain a slower heart rate to allow adequate diastolic filling across the stenotic valve, avoid tachycardia and rhythms that lose atrial kick (such as atrial fibrillation), maintain adequate preload, and avoid increases in pulmonary vascular resistance.

#160 · Organ-Based Sciences

What are the anesthetic hemodynamic goals for a patient with chronic mitral regurgitation?

Answer

Maintain a normal to slightly elevated heart rate, avoid increases in systemic vascular resistance to reduce regurgitant fraction, and maintain adequate preload and contractility, favoring forward flow into the low-pressure left atrium.

#161 • Organ-Based Sciences

What are the anesthetic hemodynamic goals for a patient with hypertrophic obstructive cardiomyopathy (HOCM)?

Answer

Maintain adequate preload, avoid tachycardia, avoid increased contractility, and maintain or increase systemic vascular resistance, since decreased preload, increased contractility, or decreased afterload worsen dynamic left ventricular outflow tract obstruction.

#162 • Organ-Based Sciences

Why does severe symptomatic aortic stenosis carry particularly high risk during noncardiac surgery?

Answer

The fixed valvular obstruction prevents compensatory increases in cardiac output during hypotension or hypovolemia, and coronary perfusion pressure can fall critically, predisposing to myocardial ischemia, arrhythmia, and cardiovascular collapse with even modest anesthetic-induced vasodilation or blood loss.

#163 • Organ-Based Sciences

How should chronic oral anticoagulation be managed perioperatively in a patient with atrial fibrillation undergoing elective surgery?

Answer

Warfarin is typically held 5 days before surgery, with bridging anticoagulation reserved for patients at high thromboembolic risk (such as a high CHA₂DS₂-VASc score or mechanical valve); direct oral anticoagulants are generally held for a shorter, drug- and renal-function-dependent interval without routine bridging.

#164 • Organ-Based Sciences

What does the CHA₂DS₂-VASc score estimate, and what are its components?

Answer

It estimates stroke risk in patients with atrial fibrillation, based on congestive heart failure, hypertension, age 75 or older (2 points), diabetes, prior stroke or TIA (2 points), vascular disease, age 65 to 74, and female sex; higher scores support anticoagulation and inform perioperative bridging decisions.

#165 · Organ-Based Sciences

What effect does placing a magnet over a permanent pacemaker typically have intraoperatively?

Answer

It converts most pacemakers to an asynchronous pacing mode (such as DOO or VOO) at a fixed rate, preventing electrocautery-induced electromagnetic interference from being misinterpreted as intrinsic cardiac activity and inappropriately inhibiting pacing output.

#166 · Organ-Based Sciences

What effect does placing a magnet over an implantable cardioverter-defibrillator (ICD) intraoperatively have?

Answer

It suspends antitachycardia therapy (shock and antitachycardia pacing) without changing the underlying pacing mode; this prevents inappropriate shocks from electrocautery interference but requires external defibrillation to be immediately available.

#167 · Organ-Based Sciences

What does the first letter of the NBG pacemaker code designate?

Answer

The chamber paced (A for atrium, V for ventricle, D for dual/both), with the second letter designating the chamber sensed, the third the response to sensing, and the fourth indicating rate modulation.

#168 · Organ-Based Sciences

What is Beck's triad, and what condition does it classically indicate?

Answer

Hypotension, jugular venous distension, and muffled heart sounds; this triad classically suggests acute cardiac tamponade, though it is often incomplete in clinical practice.

#169 · Organ-Based Sciences

What is pulsus paradoxus and how is it measured?

Answer

An exaggerated inspiratory decrease in systolic blood pressure of more than 10 mmHg; it is measured by identifying the systolic pressure at which Korotkoff sounds are first heard only during expiration versus throughout the respiratory cycle, and is a classic finding in cardiac tamponade.

#170 · Organ-Based Sciences

What key anesthetic principles guide induction in a patient with cardiac tamponade?

Answer

Maintain spontaneous ventilation and adequate preload, heart rate, and contractility until the pericardium is drained, since positive-pressure ventilation and vasodilating or negative inotropic induction agents can precipitate acute cardiovascular collapse; ketamine is often preferred for induction.

#171 · Organ-Based Sciences

How does constrictive pericarditis differ hemodynamically from cardiac tamponade?

Answer

Both impair diastolic filling and equalize diastolic pressures across chambers, but constrictive pericarditis results from a rigid, fibrotic pericardium producing a chronic, fixed restriction with a characteristic dip-and-plateau pressure waveform, whereas tamponade results from acute or subacute fluid accumulation raising intrapericardial pressure.

#172 · Organ-Based Sciences

What anesthetic drugs should be avoided in a patient with Wolff-Parkinson-White syndrome who develops atrial fibrillation?

Answer

AV nodal blocking agents such as adenosine, beta blockers, calcium channel blockers, and digoxin should be avoided, since blocking the AV node can preferentially conduct impulses down the accessory pathway and precipitate rapid ventricular response or degeneration to ventricular fibrillation.

#173 · Organ-Based Sciences

Which commonly used anesthetic and antiemetic drugs should be avoided or used cautiously in a patient with congenital long QT syndrome?

Answer

Drugs that prolong the QT interval should be avoided or minimized, including droperidol, ondansetron in high doses, volatile anesthetics with caution, and methadone; electrolyte abnormalities (hypokalemia, hypomagnesemia) should be corrected and sympathetic stimulation minimized.

#174 · Organ-Based Sciences

What is the first-line treatment for torsades de pointes?

Answer

Intravenous magnesium sulfate, regardless of serum magnesium level, along with correction of any precipitating electrolyte abnormality and discontinuation of QT-prolonging drugs; unstable or sustained torsades requires immediate defibrillation.

#175 · Organ-Based Sciences

How is perioperative myocardial infarction typically defined and detected?

Answer

It is generally defined by a rise and fall of cardiac troponin above the 99th percentile with at least one supporting feature of ischemia (symptoms, ECG changes, imaging evidence, or new wall motion abnormality); because most perioperative MIs are silent, protocols often use routine postoperative troponin surveillance in at-risk patients.

#176 · Organ-Based Sciences

What clinical outcome was the Revised Cardiac Risk Index originally validated to predict?

Answer

Major perioperative cardiac complications after noncardiac surgery, including myocardial infarction, pulmonary edema, ventricular fibrillation or cardiac arrest, and complete heart block.

#177 • Organ-Based Sciences

What echocardiographic finding characterizes grade I (impaired relaxation) diastolic dysfunction?

Answer

A reduced early mitral inflow velocity (E) relative to late atrial filling velocity (A), giving an E/A ratio less than 1, reflecting delayed left ventricular relaxation without significantly elevated filling pressures.

#178 • Organ-Based Sciences

What are key anesthetic considerations for a patient with acute right heart failure?

Answer

Avoid factors that increase pulmonary vascular resistance such as hypoxia, hypercarbia, and acidosis; maintain adequate right ventricular preload and coronary perfusion pressure; avoid excessive positive-pressure ventilation; and support contractility with inotropes such as milrinone or dobutamine while avoiding sudden increases in afterload.

#179 · Organ-Based Sciences

How does significant carotid artery stenosis affect perioperative stroke risk during noncardiac surgery?

Answer

Severe carotid stenosis, particularly if symptomatic, increases the risk of perioperative stroke due to impaired cerebral autoregulation and reduced collateral flow, so intraoperative blood pressure is typically maintained at or above the patient's baseline to preserve cerebral perfusion.

#180 · Organ-Based Sciences

Why is a slower heart rate generally beneficial in patients with significant coronary artery disease?

Answer

A slower heart rate prolongs diastole, during which the majority of coronary perfusion occurs, and reduces myocardial oxygen demand, improving the balance between myocardial oxygen supply and demand in the presence of fixed coronary stenoses.

#181 • Organ-Based Sciences

What patient factors increase the risk of pulmonary aspiration during anesthetic induction?

Answer

Full stomach or inadequate fasting, pregnancy, obesity, gastroparesis (including diabetic gastroparesis), bowel obstruction, hiatal hernia with reflux, emergency surgery, and difficult airway anticipation all increase aspiration risk and typically warrant a rapid sequence induction.

#182 • Organ-Based Sciences

What defines a rapid sequence induction (RSI) technique?

Answer

Preoxygenation followed by administration of a predetermined dose of induction agent and a rapidly acting neuromuscular blocker (such as succinylcholine or high-dose rocuronium) without intervening positive-pressure mask ventilation, aiming to secure the airway quickly and minimize the time the airway is unprotected in a patient at aspiration risk.

#183 • Organ-Based Sciences

What is the purpose of cricoid pressure (Sellick maneuver) during rapid sequence induction, and what is the current evidence regarding its efficacy?

Answer

Cricoid pressure is intended to compress the esophagus against the cervical vertebrae to prevent passive regurgitation of gastric contents into the pharynx; however, evidence supporting its efficacy is limited, it may worsen laryngoscopic view, and its routine use has become more selective and provider-dependent.

#184 • Organ-Based Sciences

What laboratory and clinical variables make up the Child-Pugh score?

Answer

Total bilirubin, serum albumin, prothrombin time (or INR), presence and severity of ascites, and presence and severity of hepatic encephalopathy; each parameter is scored 1 to 3 points and summed to classify liver disease severity as class A, B, or C.

#185 • Organ-Based Sciences

How does Child-Pugh class correlate with perioperative surgical risk in cirrhotic patients?

Answer

Child-Pugh class A carries the lowest perioperative mortality risk, class B carries intermediate risk, and class C carries the highest risk, with class C patients generally considered poor candidates for elective surgery due to markedly increased morbidity and mortality.

#186 • Organ-Based Sciences

What variables make up the MELD score, and what is it primarily used for?

Answer

The MELD score uses serum creatinine, total bilirubin, INR, and serum sodium (in the MELD-Na variant) to estimate 90-day mortality in patients with end-stage liver disease; it is used to prioritize candidates for liver transplantation and to help predict perioperative risk in cirrhotic patients undergoing other surgery.

#187 • Organ-Based Sciences

Why do patients with advanced cirrhosis develop coagulopathy despite normal or elevated fibrinogen in some cases?

Answer

The cirrhotic liver has reduced synthesis of most procoagulant clotting factors (particularly factors II, V, VII, IX, and X) but also reduced synthesis of natural anticoagulants such as protein C and protein S, producing a rebalanced but fragile hemostatic state that can shift toward bleeding or thrombosis with minimal provocation.

#188 • Organ-Based Sciences

Which clotting factors are vitamin K dependent, and why is this clinically relevant in liver disease?

Answer

Factors II, VII, IX, and X (along with proteins C and S) require vitamin K-dependent gamma-carboxylation for activity; impaired hepatic synthetic function or vitamin K malabsorption from cholestasis reduces production of these factors, prolonging the INR and contributing to bleeding risk.

#189 · Organ-Based Sciences

What is the underlying mechanism of carcinoid syndrome?

Answer

Neuroendocrine tumors, most commonly of midgut origin with hepatic metastases, secrete vasoactive mediators such as serotonin, histamine, kallikrein, and bradykinin directly into the systemic circulation, bypassing hepatic first-pass metabolism and producing flushing, diarrhea, bronchospasm, and right-sided valvular heart disease.

#190 · Organ-Based Sciences

What perioperative triggers can precipitate a carcinoid crisis, and how is it prevented?

Answer

Tumor manipulation, anxiety, hypotension, catecholamine administration, and histamine-releasing drugs can trigger massive mediator release causing severe hypotension or hypertension, bronchospasm, and arrhythmias; prophylactic octreotide infusion started before induction and continued through the procedure is the primary preventive strategy.

#191 · Organ-Based Sciences

Which anesthetic drugs should be avoided in a patient with carcinoid syndrome?

Answer

Histamine-releasing agents such as morphine, atracurium, and succinylcholine, as well as exogenous catecholamines like epinephrine and ephedrine, should be avoided, since they can provoke mediator release or worsen hemodynamic instability; hypotension is instead treated with vasopressin or octreotide.

#192 · Organ-Based Sciences

What is hepatorenal syndrome and what characterizes it?

Answer

Hepatorenal syndrome is progressive functional renal failure occurring in patients with advanced cirrhosis and portal hypertension, caused by severe splanchnic vasodilation leading to renal vasoconstriction and hypoperfusion, in the absence of other identifiable causes of kidney injury; it is often precipitated by infection, GI bleeding, or excessive diuresis.

#193 · Organ-Based Sciences

What is hepatopulmonary syndrome and how does it present?

Answer

Hepatopulmonary syndrome is intrapulmonary vascular dilation occurring in the setting of chronic liver disease that causes ventilation-perfusion mismatch and intrapulmonary shunting, presenting with hypoxemia that characteristically worsens in the upright position (platypnea-orthodeoxia).

#194 · Organ-Based Sciences

Why is succinylcholine sometimes used cautiously or avoided in patients with severe cirrhosis?

Answer

Advanced liver disease can reduce hepatic synthesis of plasma cholinesterase (pseudocholinesterase), the enzyme that metabolizes succinylcholine, which can prolong its duration of action and neuromuscular blockade.

#195 · Organ-Based Sciences

How should neuromuscular blocking agent choice be adjusted in a patient with significant hepatic dysfunction?

Answer

Agents with hepatic-independent elimination, such as cisatracurium (which undergoes Hofmann elimination and ester hydrolysis), are preferred over agents like rocuronium or vecuronium that rely substantially on hepatic metabolism and biliary excretion, since impaired liver function can prolong the duration of action of the latter.

#196 · Organ-Based Sciences

What urine output and creatinine criteria define KDIGO stage 1 acute kidney injury?

Answer

Serum creatinine increase of 0.3 mg/dL or more within 48 hours, or 1.5 to 1.9 times baseline, or urine output less than 0.5 mL/kg/hr for 6 to 12 hours.

#197 · Organ-Based Sciences

Why does chronic kidney disease prolong the action of vecuronium and rocuronium?

Answer

Both are partially cleared by renal excretion, so reduced renal clearance prolongs their duration; rocuronium is affected less than vecuronium but repeat dosing still accumulates in dialysis-dependent patients.

#198 · Organ-Based Sciences

What ECG changes progress with worsening hyperkalemia?

Answer

Peaked T waves appear first, followed by PR prolongation and P wave flattening, then QRS widening, and finally a sine wave pattern preceding ventricular fibrillation or asystole.

#199 · Organ-Based Sciences

What is the first drug given for severe hyperkalemia with ECG changes, and why?

Answer

Intravenous calcium chloride or calcium gluconate is given first because it stabilizes the cardiac cell membrane and reduces arrhythmia risk within minutes, though it does not lower serum potassium.

#200 · Organ-Based Sciences

How do insulin and glucose lower serum potassium acutely?

Answer

Insulin drives potassium intracellularly via the Na-K-ATPase pump; glucose is co-administered to prevent hypoglycemia, and the effect lasts several hours while definitive potassium removal is arranged.

#201 · Organ-Based Sciences

What classic triad defines TURP syndrome?

Answer

Hyponatremia, fluid overload, and hypoosmolality from absorption of large volumes of hypotonic irrigation fluid through open prostatic venous sinuses during transurethral resection of the prostate.

#202 · Organ-Based Sciences

Why does glycine-based irrigation fluid cause visual disturbances in TURP syndrome?

Answer

Glycine is metabolized to ammonia and glyoxylic acid, and ammonia accumulation can cause transient encephalopathy and visual disturbances including blindness, independent of the hyponatremia.

#203 • Organ-Based Sciences

How should severe symptomatic hyponatremia from TURP syndrome be corrected, and what complication must be avoided?

Answer

Hypertonic 3% saline is given to raise sodium enough to control seizures or arrhythmias, but correction should not exceed roughly 8 to 10 mEq/L in 24 hours to avoid osmotic demyelination syndrome.

#204 • Organ-Based Sciences

What preoperative hemoglobin threshold generally triggers transfusion in a stable adult without cardiac disease?

Answer

A restrictive transfusion threshold of hemoglobin below 7 g/dL is generally used in stable patients, based on trials such as TRICC showing outcomes are not worse than a liberal strategy.

#205 · Organ-Based Sciences

One unit of packed red blood cells raises hemoglobin and hematocrit by roughly how much in an average adult?

Answer

Approximately 1 g/dL of hemoglobin and 3% hematocrit in a 70 kg adult without ongoing bleeding.

#206 · Organ-Based Sciences

What is the recommended ratio of plasma, platelets, and red cells in massive transfusion protocols?

Answer

A roughly 1:1:1 ratio of fresh frozen plasma, platelets, and packed red blood cells, mimicking whole blood composition and reducing dilutional coagulopathy.

#207 · Organ-Based Sciences

What triad defines the lethal triad of trauma resuscitation?

Answer

Hypothermia, acidosis, and coagulopathy, which potentiate each other and worsen ongoing hemorrhage if not corrected together.

#208 · Organ-Based Sciences

What is the mechanism of TRALI and its typical onset?

Answer

Transfusion-related acute lung injury is caused by donor antileukocyte antibodies activating recipient neutrophils in the pulmonary vasculature, producing noncardiogenic pulmonary edema typically within 6 hours of transfusion.

#209 • Organ-Based Sciences

How is TRALI distinguished from TACO on clinical grounds?

Answer

TRALI presents with hypotension, fever, and normal or low central pressures with a noncardiogenic edema picture, whereas TACO presents with hypertension, elevated BNP, and signs of volume overload from circulatory fluid excess.

#210 • Organ-Based Sciences

What preoperative preparation is standard for a patient with sickle cell disease undergoing major surgery?

Answer

Simple or exchange transfusion to reduce hemoglobin S below about 30 percent of total hemoglobin, along with adequate hydration, warming, oxygenation, and avoidance of acidosis and stasis to reduce sickling risk.

#211 · Organ-Based Sciences

Why should tourniquets be used cautiously in patients with sickle cell disease?

Answer

Tourniquet-induced stasis, hypoxia, and acidosis in the limb promote local red cell sickling and can precipitate a vaso-occlusive crisis or compartment complications.

#212 · Organ-Based Sciences

How many days before elective surgery should clopidogrel typically be discontinued?

Answer

Clopidogrel is typically held for 5 to 7 days before elective surgery to allow platelet function to recover, since it irreversibly inhibits the P2Y₁₂ receptor for the platelet's lifespan.

#213 • Organ-Based Sciences

What is the recommended timing for stopping apixaban or rivaroxaban before a neuraxial procedure in a patient with normal renal function?

Answer

These direct factor Xa inhibitors are generally held for at least 72 hours before neuraxial anesthesia per ASRA guidance, longer with renal impairment.

#214 • Organ-Based Sciences

How does dabigatran reversal differ from that of rivaroxaban or apixaban?

Answer

Dabigatran, a direct thrombin inhibitor, is reversed with idarucizumab, a specific monoclonal antibody fragment, while the factor Xa inhibitors rivaroxaban and apixaban are reversed with andexanet alfa or four-factor prothrombin complex concentrate.

#215 • Organ-Based Sciences

What laboratory finding is characteristic of heparin-induced thrombocytopenia and how is the diagnosis supported?

Answer

A platelet count fall of more than 50 percent typically 5 to 10 days after heparin exposure, supported by a positive PF4-heparin antibody assay and, when needed, a functional serotonin release assay.

#216 • Organ-Based Sciences

Why is heparin contraindicated once heparin-induced thrombocytopenia is diagnosed, even for cardiopulmonary bypass?

Answer

Continued heparin exposure perpetuates antibody-mediated platelet activation and thrombosis risk, so alternative anticoagulants such as bivalirudin or argatroban are used instead.

#217 · Organ-Based Sciences

What ROTEM or TEG pattern suggests fibrinolysis, and what is the treatment?

Answer

A shortened clot lysis time with a rapidly decreasing amplitude after maximum clot firmness suggests hyperfibrinolysis, treated with an antifibrinolytic such as tranexamic acid.

#218 · Organ-Based Sciences

What perioperative blood glucose range is generally targeted in most hospitalized surgical patients?

Answer

A moderate target of roughly 140 to 180 mg/dL is generally recommended, avoiding both severe hyperglycemia and the harms of tight glycemic control with hypoglycemia risk.

#219 · Organ-Based Sciences

What are the diagnostic criteria for diabetic ketoacidosis?

Answer

Hyperglycemia generally above 250 mg/dL, metabolic acidosis with a low serum bicarbonate and low pH, an elevated anion gap, and positive serum or urine ketones.

#220 · Organ-Based Sciences

Why is potassium repletion started before or with insulin therapy in diabetic ketoacidosis if potassium is not elevated?

Answer

Insulin drives potassium intracellularly and can precipitate severe hypokalemia and arrhythmias in a total-body potassium-depleted DKA patient, so potassium is checked and repleted before or alongside insulin unless it is already high.

#221 · Organ-Based Sciences

How should metformin be managed before elective surgery in a patient with normal renal function?

Answer

Metformin is typically continued up to the day of surgery in patients with normal renal function, though many protocols hold it if a large contrast load or expected renal insult is anticipated, due to lactic acidosis risk with accumulation.

#222 · Organ-Based Sciences

What are the classic signs of thyroid storm under anesthesia?

Answer

Tachycardia out of proportion to anesthetic depth, hyperthermia, hypertension followed by cardiovascular collapse, and altered mental status, which can mimic malignant hyperthermia or an inadequately treated pheochromocytoma.

#223 · Organ-Based Sciences

What is first-line pharmacologic management of thyroid storm?

Answer

Beta blockade with propranolol or esmolol to control the hyperadrenergic state, along with propylthiouracil or methimazole to block new hormone synthesis, followed by iodine to block hormone release, and corticosteroids to reduce peripheral T4 to T3 conversion.

#224 · Organ-Based Sciences

What perioperative risk does severe untreated hypothyroidism carry, and how does it present?

Answer

Myxedema coma presents with hypothermia, bradycardia, hypoventilation with CO₂ retention, hyponatremia, and hypotension, and carries significant mortality if unrecognized during anesthesia.

#225 • Organ-Based Sciences

What is the physiologic basis for stress-dose corticosteroid coverage in patients with adrenal insufficiency undergoing surgery?

Answer

Surgical stress normally triggers a marked rise in endogenous cortisol; patients with primary or secondary adrenal insufficiency, or those on chronic exogenous steroids with HPA axis suppression, cannot mount this response and risk hemodynamic collapse without supplemental corticosteroids.

#226 • Organ-Based Sciences

What are the classic features of acute adrenal crisis intraoperatively?

Answer

Refractory hypotension unresponsive to fluids and vasopressors, hyponatremia, hyperkalemia, and hypoglycemia, which typically responds rapidly to intravenous hydrocortisone.

#227 · Organ-Based Sciences

What is the recommended sequence of alpha and beta blockade preoperatively for pheochromocytoma resection?

Answer

Alpha blockade with phenoxybenzamine or a selective alpha-1 blocker is started first for at least 10 to 14 days to control hypertension, and beta blockade is added only after adequate alpha blockade to prevent unopposed alpha-mediated hypertensive crisis.

#228 · Organ-Based Sciences

Why is beta blockade never given first in a patient with pheochromocytoma?

Answer

Blocking beta-2-mediated vasodilation while leaving alpha-mediated vasoconstriction unopposed can precipitate a severe hypertensive crisis and acute heart failure.

#229 • Organ-Based Sciences

What intraoperative hemodynamic pattern is expected after ligation of the pheochromocytoma's venous drainage?

Answer

Catecholamine levels fall abruptly, often producing significant hypotension that requires rapid volume resuscitation and sometimes vasopressor support after the tumor's venous outflow is clamped.

#230 • Organ-Based Sciences

Why are nondepolarizing neuromuscular blockers relatively resistant in myasthenia gravis, while depolarizing agents show an altered response?

Answer

Reduced acetylcholine receptor density from autoantibody-mediated destruction increases sensitivity to nondepolarizing blockers, requiring lower doses, while myasthenic patients often show relative resistance to succinylcholine and need higher doses for effect.

#231 • Organ-Based Sciences

What preoperative factors predict need for postoperative ventilation after thymectomy in myasthenia gravis?

Answer

Disease duration over 6 years, coexisting pulmonary disease, pyridostigmine dose above 750 mg/day, and a preoperative vital capacity less than 2.9 L are classic predictors of postoperative ventilatory support.

#232 • Organ-Based Sciences

How is a cholinergic crisis distinguished from a myasthenic crisis at the bedside?

Answer

Both cause worsening weakness, but cholinergic crisis from excess anticholinesterase therapy is accompanied by muscarinic signs such as miosis, salivation, bradycardia, and cramping, and improves with edrophonium withdrawal rather than administration.

#233 · Organ-Based Sciences

Why is succinylcholine avoided in Duchenne muscular dystrophy?

Answer

It can trigger severe hyperkalemia from upregulated extrajunctional acetylcholine receptors and has been associated with rhabdomyolysis and cardiac arrest in these patients, in addition to a malignant hyperthermia-like reaction risk.

#234 · Organ-Based Sciences

What cardiac complication should be assessed preoperatively in Duchenne muscular dystrophy?

Answer

Dilated cardiomyopathy and conduction abnormalities are common due to dystrophin deficiency in cardiac muscle, so preoperative echocardiography and ECG are recommended given the risk of perioperative heart failure or arrhythmia.

#235 · Organ-Based Sciences

What chromosome and gene are most commonly implicated in malignant hyperthermia susceptibility?

Answer

Mutations in the RYR1 gene on chromosome 19, encoding the ryanodine receptor calcium release channel in skeletal muscle sarcoplasmic reticulum, account for the majority of confirmed malignant hyperthermia susceptibility.

#236 · Organ-Based Sciences

What is the inheritance pattern of malignant hyperthermia susceptibility, and what does this mean for family screening?

Answer

It is inherited in an autosomal dominant pattern with variable penetrance, so first-degree relatives of an affected patient are considered at risk and should be counseled about testing or trigger avoidance.

#237 · Organ-Based Sciences

What is the gold standard diagnostic test for malignant hyperthermia susceptibility?

Answer

The caffeine-halothane contracture test, performed on a fresh skeletal muscle biopsy at specialized centers, measuring abnormal muscle contracture in response to caffeine and halothane exposure.

#238 · Organ-Based Sciences

What is the mechanism of dantrolene in treating malignant hyperthermia?

Answer

Dantrolene binds the ryanodine receptor and inhibits calcium release from the sarcoplasmic reticulum, reducing the sustained muscle contraction and hypermetabolism that drive the crisis.

#239 · Organ-Based Sciences

What autonomic complication is a leading cause of death in Guillain-Barre syndrome, and why does this matter for anesthetic management?

Answer

Autonomic dysfunction causing labile blood pressure and arrhythmias is a major cause of mortality, so patients can show exaggerated hypotensive or hypertensive responses to induction agents, positioning, and vasoactive drugs.

#240 · Organ-Based Sciences

Why is succinylcholine avoided in Guillain-Barre syndrome?

Answer

Denervation-like upregulation of extrajunctional acetylcholine receptors can produce exaggerated potassium release and life-threatening hyperkalemia, similar to other denervating neuromuscular conditions.

#241 • Organ-Based Sciences

What respiratory parameter is most useful for predicting the need for mechanical ventilation in Guillain-Barre syndrome?

Answer

A forced vital capacity below approximately 15 to 20 mL/kg, or a rapidly declining trend, is a key predictor of impending respiratory failure requiring intubation.

#242 • Organ-Based Sciences

Why can regional or neuraxial anesthesia be relatively contraindicated in patients with multiple sclerosis, and what is the actual evidence-based concern?

Answer

There is theoretical concern that local anesthetic neurotoxicity may worsen demyelinated spinal cord segments, particularly with spinal anesthesia and higher local anesthetic concentrations, though epidural anesthesia is generally considered safer than spinal; the true relapse risk from anesthesia itself is not clearly established and stress or fever may be more relevant triggers.

#243 • Organ-Based Sciences

Why must perioperative hyperthermia be avoided in patients with multiple sclerosis?

Answer

Elevated body temperature can transiently worsen conduction block in demyelinated axons, a phenomenon related to Uhthoff's phenomenon, potentially unmasking or worsening neurologic symptoms.

#244 • Organ-Based Sciences

Is succinylcholine considered safe in multiple sclerosis, unlike in myopathic denervating conditions?

Answer

Succinylcholine is generally considered acceptable in multiple sclerosis because it is primarily a central nervous system demyelinating disease without the widespread extrajunctional receptor upregulation seen in peripheral denervating or myopathic conditions, though caution is still used in severe disease with significant disuse atrophy.

#245 · Organ-Based Sciences

What acid-base and electrolyte pattern is typical of chronic kidney disease affecting anesthetic planning?

Answer

A normal anion gap or high anion gap metabolic acidosis from reduced acid excretion, along with hyperkalemia, hyperphosphatemia, and hypocalcemia, all of which should be optimized and ideally addressed by dialysis timing before elective surgery.

#246 · Clinical Subspecialties

Why does a neonate desaturate faster during apnea than an adult?

Answer

Neonates have a high metabolic oxygen consumption (6 to 8 mL/kg/min versus 3 to 4 mL/kg/min in adults) combined with a low functional residual capacity relative to body weight, so oxygen reserve is depleted within seconds of apnea onset.

#247 · Clinical Subspecialties

What anatomic features make the infant airway more difficult to manage than the adult airway?

Answer

Infants have a large occiput that flexes the neck, a large tongue relative to the oral cavity, a floppy omega-shaped epiglottis, a more cephalad and anterior larynx, and a short trachea, all of which make direct laryngoscopy and mask ventilation more challenging.

#248 · Clinical Subspecialties

In infants, where is the narrowest portion of the airway located?

Answer

Modern MRI-based studies show the glottis, not the cricoid ring, is actually the narrowest point of the pediatric airway, similar to adults, though the subglottic airway at the cricoid ring remains funnel-shaped, non-expandable, and less compliant, which is why appropriately sized cuffed endotracheal tubes with a leak check are preferred in modern practice.

#249 · Clinical Subspecialties

What formula estimates uncuffed endotracheal tube size for a child older than 2 years?

Answer

Uncuffed tube internal diameter (mm) equals age in years divided by 4, plus 4; for cuffed tubes subtract 0.5 mm from this calculated size.

#250 · Clinical Subspecialties

What is Robin sequence and why does it predict a difficult airway?

Answer

Robin sequence is micrognathia, glossoptosis, and cleft palate, which causes the tongue to fall posteriorly and obstruct the airway and makes both mask ventilation and laryngoscopy difficult; awake fiberoptic intubation or videolaryngoscopy is often required.

#251 · Clinical Subspecialties

Which congenital syndromes are classically associated with a difficult pediatric airway?

Answer

Pierre Robin sequence, Treacher Collins syndrome, Goldenhar syndrome, Beckwith-Wiedemann syndrome (macroglossia), and mucopolysaccharidoses such as Hurler syndrome are classic causes of a difficult pediatric airway.

#252 · Clinical Subspecialties

How does the pharmacokinetics of inhaled anesthetics differ in neonates compared with adults?

Answer

Neonates have a higher alveolar ventilation to functional residual capacity ratio and higher cardiac output to the vessel-rich group, producing faster uptake and a more rapid rise in alveolar anesthetic concentration, so induction and emergence with volatile agents are faster.

#253 · Clinical Subspecialties

How does neonatal minimum alveolar concentration (MAC) compare to that of infants and adults?

Answer

MAC is lowest in preterm neonates, peaks in infants around 1 to 6 months of age, then gradually declines through childhood and adulthood; term neonates have a lower MAC than the infant peak.

#254 · Clinical Subspecialties

Why do neonates require lower doses of highly protein-bound drugs per unit plasma concentration effect?

Answer

Neonates have decreased plasma protein concentrations (albumin and alpha-1-acid glycoprotein) and lower drug-binding affinity, leaving a higher free fraction of protein-bound drugs such as local anesthetics and benzodiazepines, increasing sensitivity at a given total dose.

#255 · Clinical Subspecialties

How does neonatal hepatic and renal immaturity affect drug clearance?

Answer

Hepatic cytochrome P450 enzyme systems and renal glomerular filtration are immature at birth, prolonging the elimination half-life of drugs cleared by hepatic metabolism or renal excretion, so dosing intervals for agents like morphine and many antibiotics must be extended in neonates.

#256 · Clinical Subspecialties

Why is succinylcholine associated with a risk of hyperkalemic cardiac arrest specifically in young boys?

Answer

Boys with undiagnosed Duchenne or Becker muscular dystrophy have upregulated extrajunctional acetylcholine receptors, and succinylcholine can trigger massive potassium efflux causing hyperkalemic cardiac arrest, which is why succinylcholine is avoided for routine elective use in young male patients unless emergency airway rescue is needed.

#257 · Clinical Subspecialties

What is the 4-2-1 rule for calculating maintenance intravenous fluid rate in children?

Answer

Give 4 mL/kg/hr for the first 10 kg of body weight, 2 mL/kg/hr for the next 10 kg, and 1 mL/kg/hr for each kilogram above 20 kg; the sums are added together to get the total hourly maintenance rate.

#258 · Clinical Subspecialties

What is the estimated maintenance fluid rate for a 25 kg child using the 4-2-1 rule?

Answer

40 mL/hr for the first 10 kg, plus 20 mL/hr for the second 10 kg, plus 5 mL/hr for the remaining 5 kg, totaling 65 mL/hr.

#259 · Clinical Subspecialties

Why is dextrose-containing fluid often avoided as the sole intraoperative maintenance fluid in most children beyond infancy?

Answer

Most children beyond infancy have adequate glycogen stores and can develop hyperglycemia and osmotic diuresis with dextrose-containing fluids intraoperatively, so isotonic non-dextrose solutions are typically used except in neonates, infants, or children at risk for hypoglycemia.

#260 · Clinical Subspecialties

Which patient populations are at higher risk for intraoperative hypoglycemia and may need dextrose-containing maintenance fluid?

Answer

Neonates, infants younger than 6 months, children on parenteral nutrition, and children with metabolic or endocrine disorders are at higher risk of hypoglycemia and often require dextrose-containing maintenance fluids intraoperatively.

#261 · Clinical Subspecialties

What electrolyte and acid-base abnormalities are classically seen in pyloric stenosis?

Answer

Hypertrophic pyloric stenosis causes repeated vomiting of gastric contents, producing hypochloremic, hypokalemic metabolic alkalosis with paradoxical aciduria as the kidneys preferentially conserve hydrogen ions over potassium to retain sodium.

#262 · Clinical Subspecialties

Is pyloric stenosis a surgical emergency requiring immediate operation?

Answer

No, pyloric stenosis is a medical emergency but a surgical urgency, not an emergency; the infant must be adequately rehydrated and the electrolyte and acid-base abnormalities corrected before proceeding to pyloromyotomy.

#263 · Clinical Subspecialties

What laboratory values indicate an infant with pyloric stenosis is ready for surgery?

Answer

Serum chloride should be greater than 100 mEq/L, serum bicarbonate less than 30 mEq/L, and urine output adequate with normal specific gravity, confirming correction of dehydration and metabolic alkalosis before anesthesia.

#264 · Clinical Subspecialties

Why is a rapid sequence induction with cricoid pressure typically used for pyloric stenosis repair, and what modification is often made?

Answer

These infants are considered a full stomach due to repeated vomiting and delayed gastric emptying, so rapid sequence induction is standard; the stomach is often first suctioned via an orogastric tube with the infant in multiple positions to empty residual contents before induction.

#265 · Clinical Subspecialties

Why are infants with corrected pyloric stenosis at risk of postoperative apnea?

Answer

Preoperative metabolic alkalosis and cerebrospinal fluid alkalosis can persist after surgical correction and blunt the central respiratory drive to hypercapnia, predisposing these infants to postoperative central apnea even if electrolytes have normalized.

#266 · Clinical Subspecialties

What is the most common cause of severe hemorrhage during tonsillectomy requiring return to the operating room?

Answer

Post-tonsillectomy hemorrhage most commonly occurs from the tonsillar fossa, either primary within 24 hours or secondary 5 to 10 days postoperatively as the eschar separates; it is a surgical emergency due to risk of aspiration and hypovolemia.

#267 · Clinical Subspecialties

How should anesthetic management differ for a child returning to the operating room for post-tonsillectomy bleeding?

Answer

These children should be treated as a full stomach with significant swallowed blood, may be hypovolemic despite normal vital signs due to compensatory mechanisms, and require rapid sequence induction with two suction devices available and blood products ready given the risk of underestimated blood loss.

#268 · Clinical Subspecialties

What is the mechanism and typical time course of emergence delirium in pediatric patients?

Answer

Emergence delirium presents as inconsolable agitation, thrashing, and disorientation typically within the first 30 minutes after emergence from anesthesia, most commonly associated with sevoflurane and desflurane, and is thought to relate to rapid changes in cortical and subcortical activity during emergence rather than pain alone.

#269 · Clinical Subspecialties

What pharmacologic strategies reduce the incidence of emergence delirium in children?

Answer

Intraoperative dexmedetomidine, propofol at the end of the case, or a low dose of fentanyl or ketamine near emergence have all been shown to reduce the incidence of emergence delirium, particularly after sevoflurane anesthesia.

#270 · Clinical Subspecialties

What is the classic triad or early sign progression of malignant hyperthermia in a child?

Answer

Early signs include unexplained tachycardia, rising end-tidal carbon dioxide despite increased minute ventilation, and masseter muscle rigidity after succinylcholine, followed by hyperthermia, generalized rigidity, and metabolic and respiratory acidosis if untreated.

#271 · Clinical Subspecialties

Why is masseter muscle rigidity after succinylcholine in a child considered a warning sign rather than always a diagnostic feature of malignant hyperthermia?

Answer

Mild jaw stiffness after succinylcholine can be a normal pharmacologic effect, but severe trismus preventing mouth opening should prompt discontinuation of triggering agents, close monitoring for other MH signs such as rising end-tidal carbon dioxide and creatine kinase, and consideration of aborting elective surgery.

#272 · Clinical Subspecialties

What is the immediate treatment for malignant hyperthermia in a pediatric patient?

Answer

Stop all triggering agents, hyperventilate with 100% oxygen, administer dantrolene 2.5 mg/kg intravenously repeated as needed, actively cool the patient, treat hyperkalemia and arrhythmias, and monitor for disseminated intravascular coagulation and myoglobinuria.

#273 • Clinical Subspecialties

Which pediatric conditions are strongly associated with susceptibility to malignant hyperthermia?

Answer

Central core disease and King-Denborough syndrome are strongly associated with malignant hyperthermia susceptibility due to ryanodine receptor mutations, and a personal or family history of unexplained perioperative death or muscle rigidity should prompt avoidance of triggering agents.

#274 • Clinical Subspecialties

In a cyanotic congenital heart lesion with a right-to-left shunt, how is inhalational induction affected?

Answer

A right-to-left shunt slows the rise of alveolar and arterial anesthetic partial pressure because shunted deoxygenated blood dilutes the arterial concentration, delaying inhalational induction, while it speeds intravenous induction because less drug is diluted by pulmonary blood flow before reaching the systemic circulation.

#275 · Clinical Subspecialties

In a lesion with a left-to-right shunt, how is inhalational induction affected compared with a normal heart?

Answer

A left-to-right shunt has minimal effect on the speed of inhalational induction because it does not significantly reduce effective pulmonary blood flow, though very large shunts can modestly slow uptake by diluting inspired anesthetic with recirculated blood.

#276 · Clinical Subspecialties

What is the physiologic goal in anesthetic management of tetralogy of Fallot to avoid a hypercyanotic spell?

Answer

Maintain adequate preload and systemic vascular resistance, avoid increases in pulmonary vascular resistance and decreases in systemic vascular resistance, and avoid hypovolemia and light anesthesia, since worsening right-to-left shunting across the ventricular septal defect and infundibular spasm precipitates a hypercyanotic spell.

#277 · Clinical Subspecialties

How should a hypercyanotic (tet) spell be treated acutely?

Answer

Administer 100% oxygen, provide intravenous fluid bolus to increase preload, give phenylephrine to increase systemic vascular resistance and reverse the right-to-left shunt, deepen anesthesia, and consider knee-chest positioning or manual abdominal aortic compression; beta-blockade can relieve infundibular spasm.

#278 · Clinical Subspecialties

What determines whether pulmonary or systemic blood flow predominates in a child with a single ventricle physiology such as hypoplastic left heart syndrome before staged palliation?

Answer

The balance between pulmonary vascular resistance and systemic vascular resistance determines the ratio of pulmonary to systemic blood flow ($Q_p:Q_s$); anesthetic management aims to balance the two circulations, avoiding excessive oxygen and hyperventilation which lower pulmonary vascular resistance and can steal flow from the systemic circulation.

#279 · Clinical Subspecialties

Why is air in the intravenous line especially dangerous in a child with an intracardiac shunt?

Answer

A right-to-left or bidirectional shunt allows venous air to bypass the pulmonary filter and enter the systemic circulation directly, causing paradoxical air embolism to the coronary or cerebral circulation, so meticulous de-airing of all intravenous lines is essential.

#280 · Clinical Subspecialties

What are the first steps of the Neonatal Resuscitation Program algorithm immediately after delivery?

Answer

Provide warmth, position the airway, clear secretions if needed, dry the infant, and stimulate; if the infant remains apneic or heart rate is below 100 beats per minute after these steps, begin positive pressure ventilation.

#281 · Clinical Subspecialties

At what heart rate threshold should chest compressions be initiated during neonatal resuscitation despite adequate ventilation?

Answer

Chest compressions should be started if the heart rate remains below 60 beats per minute after 30 seconds of effective positive pressure ventilation that adequately expands the chest.

#282 · Clinical Subspecialties

What is the compression-to-ventilation ratio used during neonatal resuscitation?

Answer

A ratio of 3 compressions to 1 ventilation (3:1) is used in neonatal resuscitation, giving approximately 90 compressions and 30 breaths per minute, unless the arrest is known to be of cardiac origin.

#283 · Clinical Subspecialties

What initial fraction of inspired oxygen is recommended when starting positive pressure ventilation in a term newborn during resuscitation?

Answer

Resuscitation of term newborns should begin with room air (21% oxygen), titrating up based on pulse oximetry targets that rise gradually over the first 10 minutes of life, since hyperoxia is associated with worse outcomes.

#284 · Clinical Subspecialties

What defines a former premature infant to be at risk for postoperative apnea after general anesthesia?

Answer

Infants born before 37 weeks gestation who are younger than 60 weeks postconceptional age (postmenstrual age) at the time of surgery are considered at increased risk for postoperative apnea and require extended monitoring.

#285 • Clinical Subspecialties

What is the recommended minimum postoperative apnea monitoring period for an at-risk former premature infant undergoing general anesthesia?

Answer

At-risk former premature infants should be monitored with continuous pulse oximetry and apnea monitoring for a minimum of 12 to 24 hours after surgery, since risk decreases but does not disappear with increasing postconceptional age.

#286 • Clinical Subspecialties

What is the presumed mechanism of postoperative apnea in former premature infants?

Answer

Immature central respiratory control centers in the brainstem produce a blunted response to hypercapnia and hypoxia, and residual anesthetic and opioid effects on this immature system increase the likelihood of central, obstructive, or mixed apnea after anesthesia.

#287 · Clinical Subspecialties

Which comorbidities further increase apnea risk in former premature infants beyond gestational and postconceptional age alone?

Answer

A history of anemia, chronic lung disease of prematurity (bronchopulmonary dysplasia), ongoing apnea of prematurity, and neurologic abnormalities all independently increase the risk of postoperative apnea in former premature infants.

#288 · Clinical Subspecialties

What is the target hemoglobin threshold below which former premature infants are considered at higher risk of postoperative apnea?

Answer

A hemoglobin concentration below 10 g/dL is associated with an increased risk of postoperative apnea in former premature infants, and correction of anemia preoperatively may reduce this risk when surgery can be delayed.

#289 · Clinical Subspecialties

Why is caffeine sometimes given prophylactically to former premature infants undergoing general anesthesia?

Answer

Caffeine citrate is a respiratory stimulant that reduces the frequency of postoperative apnea episodes in former premature infants by increasing central respiratory drive and chemoreceptor sensitivity to carbon dioxide, and it is commonly given intraoperatively in high-risk cases.

#290 · Clinical Subspecialties

How does neonatal cardiac physiology differ from the adult heart in terms of contractile reserve?

Answer

The neonatal myocardium has fewer contractile elements, less compliant ventricles, and immature calcium handling, so cardiac output is more rate-dependent than in adults, meaning bradycardia is poorly tolerated and stroke volume cannot increase substantially to compensate.

#291 • Clinical Subspecialties

Why is bradycardia the most common and most dangerous arrhythmia to treat aggressively in an anesthetized infant?

Answer

Because neonatal and infant cardiac output is heavily rate-dependent with a limited ability to augment stroke volume, bradycardia (most often caused by hypoxia in this age group) can rapidly progress to cardiovascular collapse, making prompt treatment of the underlying hypoxia and administration of atropine or epinephrine essential.

#292 • Clinical Subspecialties

What is subglottic stenosis in the context of pediatric airway management, and what historically contributed to it?

Answer

Subglottic stenosis is narrowing of the airway at the cricoid level, historically associated with prolonged intubation using an oversized uncuffed endotracheal tube causing pressure injury to the tracheal mucosa; using an appropriately sized tube with an audible leak at 20 to 30 cm H₂O reduces this risk.

#293 · Clinical Subspecialties

What is the recommended leak pressure test when selecting an endotracheal tube size in a child?

Answer

An audible air leak should be present around the endotracheal tube at approximately 20 to 30 cm H₂O of peak inspiratory pressure, confirming the tube is not too large and reducing risk of subglottic mucosal injury while still allowing adequate ventilation.

#294 · Clinical Subspecialties

How does neonatal renal glomerular filtration rate compare to adult values, and what is the clinical implication for drug dosing?

Answer

Neonatal glomerular filtration rate is only about 30 to 40 percent of adult values at birth, reaching adult levels by roughly 8 to 12 months of age, so renally cleared drugs require reduced doses or extended dosing intervals in the neonatal period.

#295 • Clinical Subspecialties

What is the estimated blood volume used for calculating allowable blood loss in a neonate versus an older child?

Answer

Estimated blood volume is approximately 90 to 100 mL/kg in a preterm neonate, 80 to 90 mL/kg in a term neonate and infant, 70 to 80 mL/kg in a child, and approximately 65 to 70 mL/kg in an adolescent or adult.

#296 • Clinical Subspecialties

Why do children require a proportionally higher volatile anesthetic concentration to reach MAC than adults, especially infants around 6 months of age?

Answer

Infants around 6 months of age have the highest MAC requirement of any age group, thought to relate to developmental differences in neural maturation and receptor density in the central nervous system, with MAC declining progressively thereafter through childhood into adulthood.

#297 · Clinical Subspecialties

What laryngoscope blade characteristics are typically preferred for neonates and young infants and why?

Answer

A straight (Miller) blade is generally preferred in neonates and young infants because it directly lifts the long, floppy epiglottis out of the line of sight, whereas the epiglottis is often too large and floppy to be effectively controlled indirectly with a curved blade placed in the vallecula.

#298 · Clinical Subspecialties

What is the appropriate approach to fasting guidelines for clear liquids in healthy children before elective anesthesia?

Answer

Current pediatric fasting guidelines allow clear liquids up to 1 hour before elective surgery in healthy children, along with 4 hours for breast milk, 6 hours for infant formula and light meals, and 8 hours for a full meal including fried or fatty foods.

#299 · Clinical Subspecialties

What ductal-dependent congenital heart lesions require prostaglandin E1 infusion to maintain patency of the ductus arteriosus before repair?

Answer

Ductal-dependent lesions such as hypoplastic left heart syndrome, critical coarctation of the aorta, interrupted aortic arch, critical pulmonary stenosis or atresia, and transposition of the great arteries often require prostaglandin E1 to keep the ductus arteriosus open and maintain systemic or pulmonary blood flow until surgical correction.

#300 · Clinical Subspecialties

What is a major side effect of prostaglandin E1 infusion in neonates that anesthesiologists must anticipate?

Answer

Prostaglandin E1 commonly causes apnea, especially in low birth weight neonates, along with fever, flushing, and hypotension, so airway support and mechanical ventilation capability should be readily available whenever a neonate is on a prostaglandin infusion.

#301 · Clinical Subspecialties

How should induction technique be modified for a child with a known or suspected difficult airway syndrome such as Treacher Collins syndrome?

Answer

Maintain spontaneous ventilation throughout induction using an inhalational technique or careful titration of intravenous agents, avoid muscle relaxants until the airway is confirmed to be manageable, and have a videolaryngoscope, fiberoptic bronchoscope, and supraglottic airway devices immediately available as backup.

#302 · Clinical Subspecialties

By how much does minute ventilation increase by term pregnancy, and what is the resulting acid-base state?

Answer

Minute ventilation increases approximately 45 to 50 percent by term, driven mainly by increased tidal volume rather than respiratory rate. This produces a compensated respiratory alkalosis with maternal PaCO₂ around 30 to 32 mmHg and serum bicarbonate around 20 mEq/L.

#303 • Clinical Subspecialties

Why do pregnant patients desaturate faster during induction of general anesthesia?

Answer

Functional residual capacity falls approximately 20 percent by term while oxygen consumption rises approximately 20 to 50 percent, sharply reducing the oxygen reservoir available during apnea. This combination shortens the safe apnea time considerably compared with nonpregnant patients.

#304 • Clinical Subspecialties

What happens to plasma volume, red cell mass, and hemoglobin concentration during normal pregnancy?

Answer

Plasma volume increases about 40 to 50 percent while red cell mass increases only about 20 to 30 percent, producing dilutional anemia of pregnancy with hemoglobin typically 11 to 12 g/dL at term. This physiologic hemodilution should not automatically be treated as iron deficiency anemia.

#305 · Clinical Subspecialties

How does cardiac output change across pregnancy, labor, and the immediate postpartum period?

Answer

Cardiac output rises about 30 to 50 percent by the third trimester through increased stroke volume and heart rate, rises further during each labor contraction from autotransfusion of uterine blood, and peaks in the first 10 to 15 minutes postpartum as caval compression is relieved and uterine blood returns to circulation. This immediate postpartum surge is a critical period for decompensation in patients with fixed cardiac lesions.

#306 · Clinical Subspecialties

What is aortocaval compression syndrome and how is it prevented?

Answer

In the supine position the gravid uterus compresses the inferior vena cava and to a lesser degree the aorta, reducing venous return and cardiac output and sometimes causing maternal hypotension, bradycardia, and fetal compromise. It is prevented by left uterine displacement, typically a 15 degree left tilt or a wedge under the right hip, maintained from mid-second trimester onward for any supine positioning.

#307 • Clinical Subspecialties

Why is gastric emptying and lower esophageal sphincter tone clinically important in pregnant patients?

Answer

Progesterone relaxes the lower esophageal sphincter and the enlarging uterus raises intragastric pressure, while gastric emptying of solids is delayed, especially once labor begins or opioids are given. All pregnant patients beyond early pregnancy are considered to have a full stomach and are managed with aspiration precautions including rapid sequence induction for general anesthesia.

#308 • Clinical Subspecialties

How does pregnancy affect minimum alveolar concentration (MAC) and local anesthetic requirements?

Answer

MAC for volatile anesthetics decreases approximately 25 to 40 percent during pregnancy, likely from elevated progesterone and endorphin activity. Epidural local anesthetic dose requirements are also reduced, partly from engorged epidural veins reducing epidural space volume and partly from increased neural sensitivity to local anesthetics.

#309 · Clinical Subspecialties

Why are pregnant patients more sensitive to local anesthetic systemic toxicity?

Answer

Increased progesterone enhances neuronal sensitivity to local anesthetics, and decreased plasma protein binding raises the free fraction of drug available to cause toxicity. Reduced functional residual capacity and increased oxygen consumption also make pregnant patients less tolerant of the hypoxemia that accompanies a toxic seizure.

#310 · Clinical Subspecialties

What anatomic and airway changes make pregnant patients more difficult to intubate?

Answer

Capillary engorgement of the airway mucosa from elevated progesterone and blood volume causes friable, edematous tissue that bleeds easily and narrows the glottic opening, and this is worsened further by preeclampsia or prolonged pushing. Weight gain, breast enlargement, and reduced neck extension also compound difficulty, so a smaller endotracheal tube and careful airway assessment are recommended for every obstetric general anesthetic.

#311 · Clinical Subspecialties

What is the pathophysiologic basis of labor pain in the first versus second stage?

Answer

First stage pain arises from uterine contraction and cervical dilation and is transmitted via visceral afferents through T10 to L1. Second stage pain adds somatic pain from vaginal and perineal distension transmitted via the pudendal nerve through S2 to S4, which is why effective second stage analgesia requires a sacral block in addition to the lumbar block.

#312 · Clinical Subspecialties

What is the standard neuraxial technique for labor analgesia and its main advantage over single-shot epidural dosing?

Answer

Combined spinal-epidural or a low-concentration epidural infusion with dilute local anesthetic plus an opioid, often using programmed intermittent epidural bolus, provides the standard approach. Programmed intermittent epidural bolus produces more consistent spread and lower total local anesthetic consumption than continuous infusion alone, improving analgesia and reducing motor block.

#313 · Clinical Subspecialties

What is the recommended dermatomal level for a cesarean delivery under neuraxial anesthesia?

Answer

A sensory block to at least T4 to T6 is required for cesarean delivery to blunt visceral peritoneal traction and prevent intraoperative pain or discomfort during uterine exteriorization. T4 corresponds to the nipple line and is the more conservative commonly cited target.

#314 · Clinical Subspecialties

How should spinal-induced hypotension during cesarean delivery be managed?

Answer

Phenylephrine is now the preferred first-line vasopressor because it restores blood pressure with less fetal acidosis than ephedrine, given as an intermittent bolus or prophylactic infusion alongside left uterine displacement and a co-load or preload of intravenous fluid. Ephedrine remains useful when significant maternal bradycardia accompanies the hypotension.

#315 • Clinical Subspecialties

Why does ephedrine cross the placenta more readily than phenylephrine, and what is the fetal consequence?

Answer

Ephedrine is a nonionized, lipophilic amine that crosses the placenta easily and stimulates fetal beta-adrenergic receptors, increasing fetal metabolic rate and oxygen consumption, which can worsen fetal acidosis in a comparably hypotensive episode. Phenylephrine crosses less readily and, by preserving maternal blood pressure and thus uteroplacental perfusion, is associated with higher fetal pH in most comparative trials.

#316 • Clinical Subspecialties

What are the diagnostic criteria for preeclampsia?

Answer

New-onset hypertension after 20 weeks gestation, defined as systolic blood pressure of at least 140 mmHg or diastolic at least 90 mmHg on two occasions, together with either proteinuria or new evidence of end-organ dysfunction such as thrombocytopenia, renal insufficiency, impaired liver function, pulmonary edema, or new cerebral or visual symptoms. Severe features include blood pressure at least 160/110 mmHg, platelets below 100,000, or the other end-organ findings listed above.

#317 · Clinical Subspecialties

What is the pathophysiology underlying preeclampsia?

Answer

Abnormal placental trophoblast invasion causes incomplete remodeling of the spiral arteries, leading to placental ischemia and release of antiangiogenic factors such as soluble fms-like tyrosine kinase-1 into the maternal circulation. These factors cause widespread endothelial dysfunction that produces vasospasm, increased vascular permeability, and the multiorgan manifestations of the disease.

#318 · Clinical Subspecialties

What is HELLP syndrome and how is it distinguished from severe preeclampsia?

Answer

HELLP syndrome stands for Hemolysis, Elevated Liver enzymes, and Low Platelets, and represents a severe variant of preeclampsia with a distinct laboratory triad rather than a separate disease. It can occur with only mild or even absent hypertension and proteinuria, so the diagnosis rests on the laboratory findings plus clinical suspicion, often with right upper quadrant pain from hepatic capsular distension.

#319 · Clinical Subspecialties

What is the drug of choice for seizure prophylaxis and treatment in preeclampsia and eclampsia, and how does it work?

Answer

Magnesium sulfate is the drug of choice, acting as an NMDA receptor antagonist and cerebral vasodilator that reduces the incidence of eclamptic seizures more effectively than phenytoin or benzodiazepines. It is typically given as a 4 to 6 gram loading dose followed by a 1 to 2 gram per hour infusion, continued through delivery and into the early postpartum period.

#320 · Clinical Subspecialties

What are the signs of magnesium toxicity and how is severe toxicity treated?

Answer

Rising serum magnesium progressively causes loss of deep tendon reflexes around 8 to 10 mEq/L, respiratory depression around 10 to 12 mEq/L, and cardiac arrest above roughly 15 mEq/L, so deep tendon reflexes and respiratory rate should be monitored during infusion. Severe toxicity is treated by stopping the infusion and administering intravenous calcium gluconate, which directly antagonizes magnesium at the neuromuscular junction.

#321 • Clinical Subspecialties

Why does magnesium sulfate potentiate both depolarizing and nondepolarizing neuromuscular blockers?

Answer

Magnesium decreases acetylcholine release at the presynaptic nerve terminal and reduces postsynaptic membrane excitability, which prolongs and intensifies neuromuscular blockade from both succinylcholine and nondepolarizing agents. Neuromuscular blocking doses should be reduced and monitored closely with a peripheral nerve stimulator in patients on magnesium infusions.

#322 • Clinical Subspecialties

What platelet count is generally considered the threshold for safely placing a neuraxial block in an obstetric patient?

Answer

A platelet count above 70,000 per microliter is generally considered acceptable for neuraxial anesthesia in an otherwise reassuring clinical picture, provided the platelet trend is stable or rising and there is no other coagulopathy. Below this level, and particularly with a rapidly falling count as in HELLP syndrome or with signs of DIC, the decision requires individualized risk-benefit assessment often supported by viscoelastic testing.

#323 · Clinical Subspecialties

Why is the trend and rate of change in platelet count as important as the absolute value when deciding on neuraxial anesthesia in preeclampsia?

Answer

A rapidly falling platelet count suggests active consumption or progression toward HELLP syndrome or DIC, so a block placed early with a reassuring but declining count can become higher risk if bleeding occurs later during a still-indwelling catheter's dwell time or at removal. Serial platelet counts should be checked in evolving preeclampsia, and the epidural catheter should be removed only when the coagulation status remains acceptable.

#324 · Clinical Subspecialties

Why should nondepolarizing neuromuscular blockers be dosed cautiously in preeclamptic patients on magnesium?

Answer

Magnesium sulfate potentiates neuromuscular blockade, so a normal induction dose of a nondepolarizing agent can produce a markedly prolonged and deeper block than intended. A reduced initial dose with quantitative train-of-four monitoring is recommended to avoid residual paralysis at extubation.

#325 · Clinical Subspecialties

What are the four categories of obstetric hemorrhage etiology summarized by the '4 T's'?

Answer

Tone (uterine atony, the most common cause), Tissue (retained placenta or products of conception), Trauma (laceration, uterine rupture, or extension of a cesarean incision), and Thrombin (coagulopathy from any cause including DIC or preexisting bleeding disorders). Systematically working through the 4 T's helps direct targeted treatment during postpartum hemorrhage.

#326 · Clinical Subspecialties

What is the first-line uterotonic for postpartum hemorrhage from uterine atony, and what are the second-line options?

Answer

Oxytocin is first-line, given as a dilute intravenous infusion rather than a rapid bolus since rapid bolus can cause significant hypotension and tachycardia. Second-line agents include methylergonovine (contraindicated in hypertension or preeclampsia), carboprost (a prostaglandin F2-alpha analogue, relatively contraindicated in asthma), and misoprostol.

#327 · Clinical Subspecialties

Why is methylergonovine avoided in a patient with preeclampsia or chronic hypertension?

Answer

Methylergonovine is a potent vasoconstrictor that can cause severe, sudden hypertension, stroke, or myocardial ischemia, which is particularly dangerous in a patient already hypertensive from preeclampsia. Carboprost or misoprostol are preferred alternatives in that setting.

#328 · Clinical Subspecialties

What is placenta accreta spectrum and why does it carry a high risk of massive hemorrhage?

Answer

Placenta accreta spectrum describes abnormal placental attachment or invasion into or through the myometrium, most often at the site of a prior cesarean scar, so the placenta cannot separate normally after delivery. Attempted manual removal causes massive uncontrolled hemorrhage, so these deliveries are planned in advance with large-bore access, blood products immediately available, and often a scheduled hysterectomy.

#329 · Clinical Subspecialties

What is uterine inversion and how is it initially managed?

Answer

Uterine inversion is turning of the uterine fundus inside out through the cervix, usually following aggressive cord traction on a placenta before it has separated, presenting with sudden severe pain, brisk hemorrhage, and vagal-mediated bradycardia or hypotension. Initial management is prompt manual replacement, often facilitated by uterine relaxation with nitroglycerin or a volatile anesthetic, before uterotonics are given once the uterus is repositioned.

#330 · Clinical Subspecialties

What is amniotic fluid embolism and how does it typically present?

Answer

Amniotic fluid embolism is a rare, unpredictable syndrome believed to result from an abnormal maternal immune response to fetal antigens entering the maternal circulation, presenting with sudden cardiovascular collapse, severe hypoxemia, and coagulopathy, often during labor or shortly after delivery. It is now understood as an anaphylactoid-like reaction rather than simple mechanical embolization of amniotic debris.

#331 · Clinical Subspecialties

What is the classic biphasic hemodynamic pattern of amniotic fluid embolism?

Answer

An initial phase of acute pulmonary hypertension and right ventricular failure causes profound hypoxemia and hypotension, followed by a second phase of left ventricular dysfunction and often disseminated intravascular coagulation with massive hemorrhage. Management is supportive, focused on cardiopulmonary resuscitation, aggressive correction of coagulopathy with blood products, and prompt delivery of the fetus if not yet delivered.

#332 · Clinical Subspecialties

What is the recommended management approach if maternal cardiac arrest occurs at a viable gestational age and resuscitation is not immediately successful?

Answer

Perimortem cesarean delivery should be initiated within about 4 minutes of arrest, with the goal of delivery by 5 minutes, since relieving aortocaval compression improves the efficacy of maternal chest compressions and can improve maternal and fetal outcomes. Delivery should not be delayed to move the patient to an operating room.

#333 • Clinical Subspecialties

What airway management modifications are recommended for general anesthesia in a pregnant patient?

Answer

Rapid sequence induction with cricoid pressure, preoxygenation to denitrogenate the reduced functional residual capacity, a smaller endotracheal tube (often 6.0 to 7.0 mm) to accommodate airway edema, and immediate availability of a difficult airway cart and video laryngoscopy are all standard modifications. Aspiration prophylaxis with a nonparticulate antacid, an H2 blocker, or metoclopramide is also routinely used.

#334 • Clinical Subspecialties

What is the recommended approach if intubation fails during cesarean delivery general anesthesia and the mother is stable?

Answer

A failed obstetric airway algorithm calls for maintaining oxygenation with bag-mask or a supraglottic airway, limiting attempts, and waking the patient if oxygenation can be maintained and there is no immediate fetal indication to proceed, since maternal safety takes priority over proceeding with an unsecured airway. If oxygenation cannot be maintained, proceeding with a supraglottic airway or surgical airway to secure oxygenation becomes necessary.

#335 • Clinical Subspecialties

What uterine tone effect do volatile anesthetics have, and why does this matter clinically?

Answer

Volatile anesthetics cause dose-dependent uterine relaxation, which is useful therapeutically for uterine inversion or internal podalic version but increases the risk of uterine atony and postpartum hemorrhage when used at higher concentrations for general anesthesia at cesarean delivery. Volatile agent concentration is typically kept low, often below 0.5 to 1 MAC, once the fetus is delivered and oxytocin is running.

#336 • Clinical Subspecialties

How does the placental transfer of most induction agents and opioids affect the neonate at cesarean delivery?

Answer

Most induction agents and opioids are lipophilic and low in molecular weight, so they cross the placenta readily by simple diffusion, meaning maternal dosing and timing directly affect neonatal drug exposure and can cause neonatal sedation or respiratory depression if given before delivery. This is why systemic opioids for labor analgesia are timed and dosed carefully relative to expected delivery.

#337 · Clinical Subspecialties

What is the mechanism and clinical significance of a 'high spinal' during obstetric anesthesia?

Answer

A high spinal occurs when local anesthetic spreads cephalad enough to block the cervical and thoracic sympathetic outflow and phrenic nerve roots, causing severe hypotension, bradycardia, and potentially apnea from diaphragmatic paralysis or brainstem involvement. Management is immediate airway support with ventilation as needed, aggressive vasopressor and fluid support, and reassurance, since the block will eventually recede.

#338 · Clinical Subspecialties

Why is a test dose used before administering the therapeutic dose of epidural local anesthetic, and what does it typically contain?

Answer

A test dose, classically 3 mL of 1.5 percent lidocaine with 1:200,000 epinephrine, is used to detect inadvertent intravascular or intrathecal catheter placement before the full therapeutic dose is given. Intravascular injection produces a transient rise in heart rate from the epinephrine, while intrathecal injection produces rapid onset of dense sensory and motor block.

#339 · Clinical Subspecialties

What is the mechanism behind the reduced local anesthetic dose requirement for spinal anesthesia during pregnancy compared with nonpregnant patients?

Answer

Engorgement of the epidural venous plexus from vena caval compression reduces the volume of the epidural and subarachnoid spaces, and elevated progesterone increases neural sensitivity to local anesthetics, both of which allow a given dose to spread further and produce a higher block than the same dose would in a nonpregnant patient. Spinal doses for cesarean delivery are therefore reduced roughly 25 to 30 percent compared with nonpregnant dosing.

#340 · Clinical Subspecialties

How should chronic anticoagulation be managed to allow safe timing of neuraxial anesthesia in a pregnant patient?

Answer

Prophylactic low molecular weight heparin requires at least a 12 hour interval before neuraxial block placement, while therapeutic dosing requires at least 24 hours, following the same consensus guidelines used in nonobstetric patients. Patients on chronic anticoagulation are often transitioned to unfractionated heparin closer to delivery because its effect is more easily monitored and reversed.

#341 · Clinical Subspecialties

What is the recommended management of a recognized accidental dural puncture with a large epidural needle during labor epidural placement?

Answer

The catheter can be threaded intrathecally and used as a continuous spinal catheter for labor analgesia, which reduces the risk of postdural puncture headache compared with re-siting the epidural, or the epidural can be resited at an adjacent interspace with careful dosing. Clear labeling of an intrathecal catheter is essential to prevent inadvertent overdose.

#342 · Clinical Subspecialties

What is the mainstay treatment for postdural puncture headache that fails conservative measures?

Answer

An epidural blood patch, typically 15 to 20 mL of the patient's own blood injected into the epidural space near the level of dural puncture, is the definitive treatment and provides durable relief in the majority of patients. Conservative measures such as hydration, caffeine, and analgesics are tried first but have limited efficacy for significant headaches.

#343 • Clinical Subspecialties

What fetal heart rate pattern is most concerning for fetal compromise and prompts urgent evaluation?

Answer

Recurrent late decelerations, indicating uteroplacental insufficiency, or a prolonged bradycardia are the most concerning patterns and often prompt immediate maternal repositioning, oxygen, fluid bolus, discontinuation of oxytocin, and consideration of expedited or emergent delivery. Category III fetal heart rate tracings, which include absent variability with recurrent decelerations or a sinusoidal pattern, are associated with significant fetal acidemia risk.

#344 • Clinical Subspecialties

Why can epidural analgesia cause a transient category II or III fetal heart rate pattern shortly after initiation?

Answer

A sudden reduction in maternal circulating catecholamines after pain relief can transiently reduce uterine tone regulation and, less commonly, maternal hypotension from sympathetic block can reduce uteroplacental perfusion, producing transient fetal bradycardia. This is usually managed with maternal repositioning, fluid bolus, and treatment of hypotension, and typically resolves without intervention.

#345 · Clinical Subspecialties

What defines a 'category 1' versus 'category 3' cesarean delivery in terms of anesthetic urgency, and how does this influence anesthetic choice?

Answer

A category 1 cesarean indicates immediate threat to maternal or fetal life requiring delivery within minutes, generally favoring rapid general anesthesia or activation of an existing well-functioning epidural, while a category 3 or elective cesarean allows time for a carefully placed spinal or epidural technique. Anesthetic technique selection always balances speed of onset against the risks of general anesthesia in the obstetric airway.

#346 · Clinical Subspecialties

How does maternal obesity compound the anesthetic risks already present in pregnancy?

Answer

Obesity further reduces functional residual capacity and increases oxygen consumption, worsens airway visualization and mask ventilation, increases technical difficulty of neuraxial block placement, and raises the risk of failed neuraxial anesthesia requiring conversion to general anesthesia. These combined factors make obese parturients a high-priority group for early anesthesia consultation and epidural placement in labor.

#347 • Clinical Subspecialties

How does aging affect volume of distribution and elimination half-life for lipophilic anesthetic drugs?

Answer

Aging increases the proportion of body fat relative to lean mass and total body water, increasing the volume of distribution for lipophilic drugs such as benzodiazepines and many opioids, which prolongs their elimination half-life even when clearance is unchanged. This means a lipophilic drug can accumulate and produce a prolonged effect despite normal organ clearance function.

#348 • Clinical Subspecialties

Why do elderly patients generally require lower induction doses of propofol and other intravenous anesthetics?

Answer

Reduced central compartment volume, decreased protein binding, slowed redistribution, and increased pharmacodynamic sensitivity of the aging brain to anesthetic agents all combine to lower the effective dose needed for loss of consciousness. Induction doses of propofol are typically reduced by roughly 30 to 50 percent in elderly patients compared with younger adults.

#349 • Clinical Subspecialties

How does aging affect minimum alveolar concentration (MAC) for volatile anesthetics?

Answer

MAC decreases approximately 6 percent per decade of age beyond young adulthood, reflecting decreased neuronal density and altered neurotransmitter function in the aging central nervous system. This age-related MAC reduction should be factored into dosing to avoid unnecessarily deep anesthesia and its associated hemodynamic and cognitive risks.

#350 • Clinical Subspecialties

What is postoperative delirium and how does it differ from postoperative cognitive dysfunction in timing and presentation?

Answer

Postoperative delirium is an acute, fluctuating disturbance of attention and awareness typically arising within the first few postoperative days, presenting as hyperactive, hypoactive, or mixed subtypes. Postoperative cognitive dysfunction is a more subtle, longer-lasting decline in memory, concentration, or executive function detected on neuropsychological testing that can persist for weeks to months after surgery.

#351 • Clinical Subspecialties

What are the major modifiable risk factors for postoperative delirium in elderly surgical patients?

Answer

Modifiable risk factors include uncontrolled pain, use of benzodiazepines and anticholinergic medications, sleep deprivation, sensory deprivation from missing hearing aids or glasses, dehydration, and undertreated hypoxemia or metabolic derangement. Multicomponent nonpharmacologic prevention protocols targeting these factors, such as early mobilization and reorientation, are the most effective preventive strategy.

#352 • Clinical Subspecialties

Does the choice of general versus neuraxial anesthesia clearly change the incidence of postoperative delirium or cognitive dysfunction in elderly patients?

Answer

Large randomized trials have generally not shown a clear reduction in postoperative delirium or long-term cognitive dysfunction with neuraxial or regional anesthesia compared with general anesthesia, so anesthetic type alone should not be viewed as the primary preventive lever. Depth of anesthesia monitoring to avoid excessively deep general anesthesia and minimizing benzodiazepines appear more consistently protective.

#353 • Clinical Subspecialties

What is frailty and why is it a better predictor of postoperative outcomes than chronological age alone?

Answer

Frailty is a clinical syndrome of decreased physiologic reserve and increased vulnerability to stressors, often assessed by criteria such as unintentional weight loss, weakness, slow gait speed, low activity, and exhaustion. Frailty predicts postoperative complications, prolonged length of stay, and mortality more accurately than chronological age because it captures functional reserve rather than years lived.

#354 • Clinical Subspecialties

Why are elderly patients particularly vulnerable to the effects of polypharmacy in the perioperative period?

Answer

Elderly patients often take multiple chronic medications with overlapping sedative, anticholinergic, or hypotensive effects, and reduced hepatic and renal clearance increases the risk of drug accumulation and drug-drug interactions with anesthetic agents. Careful preoperative medication reconciliation is important to identify agents that increase delirium risk or interact adversely with planned anesthetics.

#355 · Clinical Subspecialties

How does aging affect the cardiovascular response to hypotension and the risk of anesthetic-induced hemodynamic instability?

Answer

Decreased beta-adrenergic receptor responsiveness, reduced arterial compliance, and diminished baroreceptor sensitivity blunt the compensatory tachycardic and vasoconstrictive response to hypotension in elderly patients. This makes elderly patients more prone to profound and prolonged hypotension after induction doses of anesthetic agents that would be well tolerated in a younger patient.

#356 · Clinical Subspecialties

Why do elderly patients have prolonged recovery from neuromuscular blocking agents even at standard doses?

Answer

Decreased hepatic and renal clearance slows elimination of aminosteroid neuromuscular blockers such as rocuronium and vecuronium, and reduced muscle mass and altered receptor sensitivity can further prolong clinical effect. Quantitative neuromuscular monitoring and appropriately timed reversal are especially important in elderly patients to avoid residual postoperative paralysis.

#357 · Clinical Subspecialties

What perioperative strategy is recommended to reduce postoperative delirium related to opioid and sedative use in elderly surgical patients?

Answer

A multimodal, opioid-sparing analgesic strategy using regional techniques, acetaminophen, and nonopioid adjuncts where appropriate, combined with avoiding benzodiazepines and minimizing anticholinergic drugs, reduces sedative burden while still controlling pain. Adequate but not excessive analgesia is important because both undertreated pain and oversedation independently increase delirium risk.

#358 · Clinical Subspecialties

What is multimodal analgesia and why is it preferred for acute postoperative pain?

Answer

Multimodal analgesia combines drugs with different mechanisms (acetaminophen, NSAIDs, gabapentinoids, local anesthetics, low-dose ketamine, regional techniques) with opioids. It improves pain control while reducing total opioid dose and opioid-related side effects.

#359 · Clinical Subspecialties

Describe the WHO analgesic ladder for cancer pain management.

Answer

Step 1 uses non-opioids (acetaminophen or NSAIDs) for mild pain, step 2 adds a weak opioid for moderate pain, and step 3 uses a strong opioid such as morphine for severe pain. Adjuvant medications can be added at any step.

#360 · Clinical Subspecialties

What are the Budapest criteria used for?

Answer

The Budapest criteria are the clinical diagnostic criteria for complex regional pain syndrome, requiring continuing pain disproportionate to the inciting event plus symptoms and signs in at least three of four categories: sensory, vasomotor, sudomotor/edema, and motor/trophic.

#361 · Clinical Subspecialties

What is the mechanism underlying opioid tolerance?

Answer

Chronic opioid exposure leads to receptor downregulation, desensitization, and upregulation of the cyclic AMP pathway, requiring escalating doses to achieve the same analgesic effect over time.

#362 · Clinical Subspecialties

How does opioid-induced hyperalgesia differ clinically from opioid tolerance?

Answer

Tolerance presents as diminishing analgesia from a stable opioid dose. Opioid-induced hyperalgesia presents as worsening or diffuse pain that increases with escalating opioid doses; management favors dose reduction, opioid rotation, or NMDA antagonists rather than further dose escalation.

#363 · Clinical Subspecialties

What are first-line pharmacologic agents for neuropathic pain?

Answer

Gabapentinoids (gabapentin, pregabalin), serotonin-norepinephrine reuptake inhibitors (duloxetine), and tricyclic antidepressants (nortriptyline) are considered first-line for neuropathic pain.

#364 · Clinical Subspecialties

What is the mechanism of action of gabapentin and pregabalin?

Answer

Gabapentinoids bind the alpha-2-delta subunit of voltage-gated calcium channels on presynaptic neurons, reducing calcium influx and decreasing release of excitatory neurotransmitters involved in pain signaling.

#365 · Clinical Subspecialties

Why is low-dose ketamine useful as an adjunct in opioid-tolerant surgical patients?

Answer

Ketamine is an NMDA receptor antagonist that blocks central sensitization and can reverse or prevent opioid-induced hyperalgesia, providing opioid-sparing analgesia in patients with high baseline opioid requirements.

#366 · Clinical Subspecialties

What is the primary indication for a celiac plexus block?

Answer

Celiac plexus block or neurolysis is indicated for visceral pain from upper abdominal malignancies, most commonly pancreatic cancer, providing significant opioid-sparing analgesia.

#367 · Clinical Subspecialties

What condition is a stellate ganglion block used to treat?

Answer

Stellate ganglion block targets the cervicothoracic sympathetic chain and is used for sympathetically maintained pain conditions such as complex regional pain syndrome of the upper extremity and refractory upper extremity vascular insufficiency.

#368 · Clinical Subspecialties

When is spinal cord stimulation considered for chronic pain?

Answer

Spinal cord stimulation is considered for chronic neuropathic pain refractory to conservative therapy, such as failed back surgery syndrome or complex regional pain syndrome, after a successful trial period demonstrates meaningful pain relief.

#369 · Clinical Subspecialties

What is the role of an intrathecal drug delivery pump in cancer pain management?

Answer

Intrathecal pumps deliver opioids and adjuvants directly to the cerebrospinal fluid, allowing effective analgesia at a fraction of the systemic dose, which reduces systemic side effects for patients with refractory cancer pain and limited life expectancy.

#370 · Clinical Subspecialties

What pharmacologic properties make methadone unique among opioids?

Answer

Methadone is a mu-opioid agonist with additional NMDA receptor antagonism and monoamine reuptake inhibition, giving it efficacy against neuropathic and opioid-tolerant pain. It has a long and variable half-life and can prolong the QTc interval, requiring careful titration and ECG monitoring.

#371 · Clinical Subspecialties

What is the mechanism of buprenorphine and why does it have a ceiling effect?

Answer

Buprenorphine is a partial mu-opioid agonist with high receptor affinity and slow dissociation. Its partial agonism produces a plateau in respiratory depression at higher doses, giving it a favorable safety profile compared with full agonists, while it retains meaningful dose-dependent analgesic efficacy across the typical clinical dosing range.

#372 · Clinical Subspecialties

How should chronic buprenorphine therapy be managed perioperatively for a patient undergoing major surgery?

Answer

Current practice favors continuing buprenorphine perioperatively in most cases, supplementing with full opioid agonists and multimodal analgesia as needed, since buprenorphine's high receptor affinity does not preclude adequate analgesia and discontinuation risks relapse.

#373 · Clinical Subspecialties

What are the three FDA-approved medications for medication-assisted treatment of opioid use disorder?

Answer

Methadone (full agonist), buprenorphine (partial agonist), and naltrexone (antagonist) are the three approved medications for opioid use disorder treatment, each requiring different induction protocols and monitoring.

#374 · Clinical Subspecialties

What is the mechanism of naloxone and its onset relative to typical opioid duration?

Answer

Naloxone is a competitive mu-opioid receptor antagonist that rapidly reverses opioid-induced respiratory depression, but its duration of action is shorter than most opioids, so repeat dosing or infusion may be needed to prevent renarcotization.

#375 · Clinical Subspecialties

What is the general treatment approach for complex regional pain syndrome?

Answer

Treatment emphasizes early functional restoration through physical and occupational therapy, combined with pharmacologic pain control (gabapentinoids, tricyclics, corticosteroids in early inflammatory stage), and sympathetic blocks or spinal cord stimulation for refractory cases.

#376 · Clinical Subspecialties

What is the preferred pharmacologic treatment for postherpetic neuralgia?

Answer

First-line treatments include gabapentin, pregabalin, tricyclic antidepressants, and topical lidocaine patches, with topical capsaicin as an additional option for localized pain.

#377 • Clinical Subspecialties

What procedure is used to treat chronic facet joint mediated back pain after a positive diagnostic block?

Answer

Radiofrequency ablation of the medial branch nerves supplying the facet joint is used to provide longer-lasting relief after a diagnostic medial branch block confirms the pain generator.

#378 • Clinical Subspecialties

What is the rationale for epidural steroid injections in the treatment of radicular pain?

Answer

Epidural steroid injections deliver corticosteroid near an inflamed or compressed nerve root to reduce local inflammation and edema, providing temporary relief of radicular pain, most often from disc herniation or spinal stenosis.

#379 · Clinical Subspecialties

What are the core components of an opioid-sparing multimodal perioperative pain regimen?

Answer

A typical regimen includes scheduled acetaminophen, an NSAID or COX-2 inhibitor if not contraindicated, a gabapentinoid, regional or neuraxial anesthesia when feasible, and opioids reserved for breakthrough pain.

#380 · Clinical Subspecialties

What are the tapering principles for a chronic opioid patient discontinuing therapy?

Answer

Opioid tapers should be gradual and individualized, typically reducing the dose by 10 percent every one to four weeks, with close monitoring for withdrawal symptoms and psychological support to avoid destabilizing the patient.

#381 · Clinical Subspecialties

What are the Sepsis-3 definitions of sepsis and septic shock?

Answer

Sepsis is life-threatening organ dysfunction caused by a dysregulated host response to infection, identified by an increase in SOFA score of 2 or more points. Septic shock is a subset of sepsis with persistent hypotension requiring vasopressors to maintain a mean arterial pressure of 65 mmHg or greater, plus a serum lactate above 2 mmol/L despite adequate fluid resuscitation.

#382 · Clinical Subspecialties

What are the components of the Surviving Sepsis Campaign hour-1 bundle?

Answer

The hour-1 bundle includes measuring lactate, obtaining blood cultures before antibiotics, administering broad-spectrum antibiotics, beginning rapid administration of 30 mL/kg crystalloid for hypotension or lactate above 4 mmol/L, and starting vasopressors if the patient remains hypotensive during or after fluid resuscitation.

#383 · Clinical Subspecialties

What is the qSOFA score and how is it used?

Answer

The quick SOFA score uses respiratory rate 22 or higher, altered mentation, and systolic blood pressure 100 mmHg or lower, each worth one point. A score of 2 or more outside the ICU suggests higher risk of poor outcome and should prompt further sepsis workup.

#384 · Clinical Subspecialties

What is the recommended initial fluid choice and volume for sepsis resuscitation?

Answer

Balanced crystalloid solutions are recommended as first-line fluid, with an initial bolus of 30 mL/kg within the first three hours for sepsis-induced hypoperfusion, followed by dynamic reassessment of fluid responsiveness.

#385 · Clinical Subspecialties

What is the first-line vasopressor for septic shock?

Answer

Norepinephrine is the first-line vasopressor for septic shock because it provides potent alpha-1 mediated vasoconstriction with modest beta-1 inotropic support and less tachycardia than other agents.

#386 · Clinical Subspecialties

What agent is typically added when norepinephrine alone fails to achieve target MAP in septic shock?

Answer

Vasopressin is typically added as a second agent to reduce the norepinephrine dose requirement, acting through V1 receptors independent of the catecholamine pathway; epinephrine is an alternative second or third agent.

#387 · Clinical Subspecialties

What are the Berlin criteria for diagnosing ARDS?

Answer

The Berlin definition requires onset within one week of a known clinical insult, bilateral opacities on chest imaging not fully explained by effusion or collapse, respiratory failure not fully explained by cardiac failure or fluid overload, and a PaO₂/FiO₂ ratio of 300 or less with at least 5 cmH₂O of PEEP, stratified into mild, moderate, and severe categories.

#388 · Clinical Subspecialties

What tidal volume strategy defines lung-protective ventilation in ARDS?

Answer

Lung-protective ventilation uses low tidal volumes of 4 to 8 mL/kg predicted body weight, based on the ARDSNet protocol, to minimize ventilator-induced lung injury from volutrauma and barotrauma.

#389 · Clinical Subspecialties

What plateau pressure target is used in lung-protective ventilation for ARDS?

Answer

Plateau pressure should be kept at 30 cmH₂O or below to limit alveolar overdistension and reduce the risk of ventilator-induced lung injury.

#390 · Clinical Subspecialties

How is PEEP typically titrated in patients with ARDS?

Answer

PEEP is titrated using a PEEP/FiO₂ table balancing oxygenation against overdistension, or by other methods such as esophageal pressure monitoring or best compliance, aiming to recruit collapsed alveoli while avoiding excessive plateau pressure.

#391 · Clinical Subspecialties

What is the mortality benefit and indication for prone positioning in ARDS?

Answer

Prone positioning improves ventilation-perfusion matching and reduces mortality in patients with moderate to severe ARDS, defined as $\text{PaO}_2/\text{FiO}_2$ below 150, and is typically applied for at least 12 to 16 hours per session.

#392 · Clinical Subspecialties

What is driving pressure and why is it clinically important in ARDS?

Answer

Driving pressure is plateau pressure minus PEEP, reflecting tidal volume normalized to respiratory system compliance. Lower driving pressure is independently associated with improved survival in ARDS, making it a useful target beyond tidal volume alone.

#393 · Clinical Subspecialties

When is ECMO considered in the management of severe ARDS?

Answer

Venovenous ECMO is considered for severe refractory hypoxemia despite optimized lung-protective ventilation, prone positioning, and neuromuscular blockade, typically when $\text{PaO}_2/\text{FiO}_2$ remains below 80 to 100 or when severe hypercapnic acidosis persists.

#394 · Clinical Subspecialties

What are the components of the CAM-ICU assessment for delirium?

Answer

CAM-ICU requires acute onset or fluctuating course, inattention, plus either disorganized thinking or an altered level of consciousness. A positive result requires the first two features plus at least one of the last two.

#395 · Clinical Subspecialties

What are major risk factors for ICU delirium?

Answer

Risk factors include advanced age, preexisting cognitive impairment, mechanical ventilation, benzodiazepine use, sleep deprivation, immobility, and severity of critical illness.

#396 · Clinical Subspecialties

What is the recommended first-line approach to managing ICU delirium?

Answer

Nonpharmacologic measures are first-line, including reorientation, minimizing sedation, promoting normal sleep-wake cycles, early mobilization, and correcting underlying metabolic or infectious contributors; antipsychotics are reserved for severe agitation causing safety risk.

#397 · Clinical Subspecialties

Why is dexmedetomidine often preferred over benzodiazepines for ICU sedation?

Answer

Dexmedetomidine is a selective alpha-2 agonist that provides sedation and analgesia without significant respiratory depression, and it is associated with lower rates of delirium and shorter time to extubation compared with benzodiazepine-based sedation.

#398 · Clinical Subspecialties

What sedation scale is commonly used to titrate ICU sedation, and what is a typical target?

Answer

The Richmond Agitation-Sedation Scale (RASS) ranges from plus 4 (combative) to minus 5 (unarousable). A common target for most critically ill patients is light sedation, RASS 0 to minus 2, to facilitate assessment and reduce delirium.

#399 · Clinical Subspecialties

What is the purpose of paired spontaneous awakening trials and spontaneous breathing trials in the ICU?

Answer

Daily interruption of sedation paired with a spontaneous breathing trial reduces oversedation, shortens duration of mechanical ventilation, and decreases ICU length of stay by identifying patients ready for liberation from the ventilator.

#400 · Clinical Subspecialties

What does the ABCDEF bundle stand for in ICU care?

Answer

Assess, prevent, and manage pain; Both spontaneous awakening and breathing trials; Choice of analgesia and sedation; Delirium assess, prevent, and manage; Early mobility and exercise; Family engagement and empowerment. It is a bundled approach shown to improve ICU outcomes.

#401 · Clinical Subspecialties

When should enteral nutrition be initiated in critically ill patients, and why is early feeding favored?

Answer

Enteral nutrition should be started within 24 to 48 hours of ICU admission in hemodynamically stable patients, since early enteral feeding preserves gut mucosal integrity, reduces infectious complications, and is preferred over parenteral nutrition when the gut is functional.

#402 · Clinical Subspecialties

What is refeeding syndrome and which electrolyte abnormality is its hallmark?

Answer

Refeeding syndrome occurs when nutrition is reintroduced after prolonged starvation, causing an insulin surge that drives potassium, magnesium, and especially phosphate intracellularly. Severe hypophosphatemia is the hallmark abnormality and can cause cardiac and respiratory failure.

#403 · Clinical Subspecialties

What are the characteristic hemodynamic findings in distributive shock?

Answer

Distributive shock, as seen in sepsis, presents with low systemic vascular resistance, normal to high cardiac output, and low to normal filling pressures due to profound vasodilation and capillary leak.

#404 · Clinical Subspecialties

What are the characteristic hemodynamic findings in cardiogenic shock?

Answer

Cardiogenic shock presents with low cardiac output, elevated systemic vascular resistance, and elevated cardiac filling pressures, reflecting primary pump failure with compensatory vasoconstriction.

#405 · Clinical Subspecialties

What are the characteristic hemodynamic findings in hypovolemic shock?

Answer

Hypovolemic shock presents with low cardiac output, low cardiac filling pressures, and elevated systemic vascular resistance from compensatory vasoconstriction in response to reduced circulating volume.

#406 · Clinical Subspecialties

What are common causes of obstructive shock?

Answer

Obstructive shock results from mechanical obstruction to cardiac filling or output, including cardiac tamponade, tension pneumothorax, and massive pulmonary embolism, each impairing ventricular filling or forward flow.

#407 · Clinical Subspecialties

What is the mechanism of vasopressin as a vasopressor in shock states?

Answer

Vasopressin acts on V1 receptors on vascular smooth muscle to cause vasoconstriction independent of adrenergic pathways, making it useful in catecholamine-resistant vasodilatory shock and as a norepinephrine-sparing adjunct.

#408 · Clinical Subspecialties

When is epinephrine typically used as a vasopressor in shock, and what are its effects?

Answer

Epinephrine provides potent beta-1 inotropic and chronotropic effects along with alpha-1 vasoconstriction, and is used as a second or third-line agent in refractory septic shock or as first-line support in cardiogenic shock and anaphylaxis.

#409 · Clinical Subspecialties

Why is serial lactate clearance used to guide resuscitation in septic shock?

Answer

Elevated lactate reflects tissue hypoperfusion and anaerobic metabolism. Failure of lactate to clear with resuscitation is associated with worse outcomes, so serial measurements help guide the adequacy and titration of fluid and vasopressor therapy.

#410 · Clinical Subspecialties

What does a low central venous oxygen saturation (ScvO₂) suggest during shock resuscitation?

Answer

A low ScvO₂, typically below 70 percent, suggests inadequate oxygen delivery relative to tissue oxygen demand, prompting evaluation of cardiac output, hemoglobin, arterial oxygenation, and further hemodynamic optimization.

#411 · Clinical Subspecialties

When should stress-dose corticosteroids be considered in septic shock?

Answer

Low-dose hydrocortisone is considered for patients with septic shock who remain hemodynamically unstable despite adequate fluid resuscitation and vasopressor therapy, as it can reduce time to shock resolution though evidence for mortality benefit is mixed.

#412 · Clinical Subspecialties

What are key components of a ventilator-associated pneumonia prevention bundle?

Answer

The bundle typically includes elevating the head of the bed to 30 to 45 degrees, daily sedation interruption with readiness-to-extubate assessment, oral care with chlorhexidine, subglottic secretion drainage, and minimizing duration of mechanical ventilation.

#413 • Clinical Subspecialties

What are the primary indications for initiating continuous renal replacement therapy in critically ill patients with acute kidney injury?

Answer

Indications include refractory volume overload, severe hyperkalemia, severe metabolic acidosis, uremic complications, and certain drug or toxin removal, especially in hemodynamically unstable patients who cannot tolerate intermittent hemodialysis.

#414 • Clinical Subspecialties

Why is preoperative discussion of airway plan with the surgeon mandatory before shared-airway ENT cases such as microlaryngoscopy?

Answer

The anesthesiologist and surgeon share the same anatomic space, so the surgical approach (suspension laryngoscopy, jet ventilation, or a small tube) dictates the airway technique and must be agreed on before induction to avoid an intraoperative crisis.

#415 · Clinical Subspecialties

What are the two major postoperative airway complications after tonsillectomy and adenoidectomy in children with OSA?

Answer

Post-tonsillectomy hemorrhage causing airway soiling and a difficult reintubation, and opioid-induced respiratory depression, since these children have heightened sensitivity to opioids from chronic hypoxemia.

#416 · Clinical Subspecialties

Why should opioid dosing be reduced in children undergoing adenotonsillectomy for severe obstructive sleep apnea?

Answer

Chronic intermittent hypoxemia upregulates mu-opioid receptors and blunts central chemoreceptor responsiveness, making these children more sensitive to the respiratory depressant effects of opioids at standard weight-based doses.

#417 · Clinical Subspecialties

What is the recommended management of a patient presenting for surgery with a bleeding tonsil after recent tonsillectomy?

Answer

Treat as a full stomach with a rapid sequence induction, have two suction devices ready, place the patient head-down or lateral until intubation, and anticipate a difficult laryngoscopy from blood and hypovolemia; extubate only when fully awake.

#418 · Clinical Subspecialties

List the three components of the fire triad relevant to airway laser surgery.

Answer

An ignition source (the laser), an oxidizer (oxygen or nitrous oxide), and a fuel (the endotracheal tube, surgical drapes, or airway secretions).

#419 · Clinical Subspecialties

What FiO₂ should be used during laser surgery on or near the airway, and why?

Answer

The lowest FiO₂ that maintains adequate oxygenation, ideally at or below 30%, with air as the diluent instead of nitrous oxide, since both oxygen and nitrous oxide support combustion and increase airway fire risk.

#420 · Clinical Subspecialties

What is the immediate management sequence for an airway fire during laser surgery?

Answer

Stop ventilation, disconnect the breathing circuit, remove the burning endotracheal tube, pour saline into the airway to extinguish residual fire, then mask ventilate with 100% oxygen and reintubate to assess and manage thermal airway injury with bronchoscopy.

#421 • Clinical Subspecialties

Which endotracheal tube feature reduces the risk of airway fire during laser microlaryngeal surgery?

Answer

A laser-resistant tube, typically with a metallic or wrapped shaft and saline-filled cuffs (often dyed with methylene blue to signal cuff rupture), reduces the chance of tube ignition or perforation by the laser beam.

#422 • Clinical Subspecialties

Why is postoperative monitoring for airway edema especially important after surgery for obstructive sleep apnea (e.g., uvulopalatopharyngoplasty)?

Answer

Pharyngeal manipulation and tissue swelling can compromise an already anatomically narrow airway, and residual anesthetic and opioid effects further increase risk of upper airway obstruction, so extended monitoring in a high-acuity setting is warranted.

#423 · Clinical Subspecialties

What is the maximum recommended dose of lidocaine with epinephrine used in tumescent local anesthesia for liposuction?

Answer

Up to 35 to 55 mg/kg has been described as safe in tumescent technique due to slow systemic absorption from the dilute, vasoconstricted subcutaneous depot, though peak plasma levels occur 12 to 14 hours after infiltration and delayed toxicity must be monitored for.

#424 · Clinical Subspecialties

Why does tumescent lidocaine carry a risk of delayed systemic toxicity compared with standard local infiltration?

Answer

The large volume of dilute lidocaine with epinephrine is absorbed slowly from the vasoconstricted, poorly perfused adipose tissue, so peak plasma lidocaine concentrations occur 12 to 14 hours after injection, well after the patient may have left monitored care.

#425 · Clinical Subspecialties

What is the most feared systemic complication of large-volume liposuction, and what causes it?

Answer

Pulmonary fat embolism and volume shifts causing pulmonary edema; large-volume fat aspiration and fluid resuspension can precipitate fluid overload, and disrupted fat can embolize to the pulmonary circulation.

#426 · Clinical Subspecialties

What intraoperative monitoring priority is unique to free flap microvascular reconstruction anesthesia?

Answer

Maintaining flap perfusion by avoiding hypotension, hypothermia, and vasoconstriction; anesthetic goals favor generous fluid administration, normothermia, adequate analgesia, and often avoidance of vasopressors that could compromise the anastomosed vessel.

#427 · Clinical Subspecialties

Why is normothermia especially important during prolonged free flap reconstructive surgery?

Answer

Hypothermia causes peripheral vasoconstriction that can reduce blood flow to the microvascular anastomosis and flap, increasing the risk of flap thrombosis and failure.

#428 · Clinical Subspecialties

What VTE prophylaxis strategy is recommended for high-risk plastic surgery patients undergoing prolonged procedures such as abdominoplasty combined with other body contouring?

Answer

Mechanical prophylaxis with intermittent pneumatic compression intraoperatively, combined with risk-stratified pharmacologic prophylaxis (e.g., low molecular weight heparin) postoperatively per the Caprini risk assessment score, since combined procedures and prolonged supine or flexed positioning increase VTE risk.

#429 · Clinical Subspecialties

How does the Caprini risk assessment model guide perioperative VTE prophylaxis decisions in plastic surgery patients?

Answer

It assigns points for patient and procedural risk factors (age, obesity, immobility, procedure duration, prior VTE) to stratify patients into risk categories that determine whether mechanical prophylaxis alone or the addition of pharmacologic anticoagulation is indicated.

#430 · Clinical Subspecialties

What physiologic changes result from CO₂ pneumoperitoneum during laparoscopic surgery?

Answer

Increased intra-abdominal pressure raises systemic vascular resistance and mean arterial pressure, decreases venous return and cardiac output at higher pressures, and CO₂ absorption causes hypercapnia and respiratory acidosis requiring increased minute ventilation.

#431 • Clinical Subspecialties

How does pneumoperitoneum affect respiratory mechanics during laparoscopic surgery?

Answer

Elevated intra-abdominal pressure pushes the diaphragm cephalad, reducing functional residual capacity and lung compliance and increasing peak airway pressures, which can promote atelectasis, especially combined with steep Trendelenburg positioning.

#432 • Clinical Subspecialties

What is the recommended maximum intra-abdominal insufflation pressure during standard laparoscopic surgery, and why?

Answer

Generally 12 to 15 mmHg, because higher pressures progressively reduce venous return and cardiac output, increase the risk of gas embolism, and impair respiratory compliance.

#433 • Clinical Subspecialties

What cardiovascular and cerebral effects accompany steep Trendelenburg positioning during robotic pelvic surgery?

Answer

Increased venous return raises central venous pressure and can precipitate pulmonary edema in susceptible patients, while cephalad fluid shift increases cerebral venous pressure and intraocular and intracranial pressure, risking cerebral and airway edema with prolonged positioning.

#434 • Clinical Subspecialties

Why is airway edema a specific concern after prolonged robotic surgery in steep Trendelenburg position?

Answer

Sustained cephalad fluid shift over several hours can cause facial, laryngeal, and tongue edema, increasing the risk of postextubation airway obstruction, so a cuff leak test is recommended before extubation in prolonged cases.

#435 · Clinical Subspecialties

What unique intraoperative access limitation must be planned for before docking a surgical robot?

Answer

Once the robot is docked, direct access to the patient, airway, and intravenous lines is severely restricted, so all lines, monitors, and airway security must be confirmed and undocking protocols with the surgical team established before starting the case.

#436 · Clinical Subspecialties

What are the classic signs of venous CO₂ gas embolism during laparoscopic surgery?

Answer

Sudden decrease in end-tidal CO₂, hypotension, hypoxemia, a mill-wheel murmur on auscultation, and dysrhythmias; severe cases can progress to cardiovascular collapse.

#437 · Clinical Subspecialties

What is the immediate management of suspected CO₂ gas embolism during laparoscopy?

Answer

Notify the surgeon to stop insufflation and desufflate the abdomen, place the patient in left lateral decubitus with head-down (Durant position) to trap air in the right ventricular apex away from the outflow tract, give 100% oxygen, and support hemodynamics with fluids and vasopressors as needed.

#438 · Clinical Subspecialties

Why does venous CO₂ embolism tend to be better tolerated than air embolism of similar volume?

Answer

CO₂ is far more soluble in blood than air, allowing rapid absorption and reducing the volume of gas that persists to obstruct right ventricular outflow, though large or rapid boluses can still cause hemodynamic collapse.

#439 · Clinical Subspecialties

What is the oculocardiac reflex and what triggers it?

Answer

A trigeminovagal reflex causing bradycardia, junctional rhythm, or asystole in response to traction on extraocular muscles, pressure on the globe, or ocular manipulation, most commonly seen with strabismus surgery.

#440 · Clinical Subspecialties

What is the first step in managing bradycardia from the oculocardiac reflex during strabismus surgery?

Answer

Immediately ask the surgeon to stop the stimulus (release traction on the muscle), which typically resolves the bradycardia; if it persists or recurs, give an anticholinergic such as atropine or glycopyrrolate.

#441 • Clinical Subspecialties

What serious complication of retrobulbar block can cause immediate loss of consciousness and apnea?

Answer

Brainstem anesthesia from inadvertent injection into the optic nerve sheath, allowing local anesthetic to track along the subarachnoid space to the brainstem, causing contralateral mydriasis, loss of consciousness, apnea, and hemodynamic instability requiring supportive ventilation until it resolves.

#442 • Clinical Subspecialties

What globe injury risk is specifically associated with retrobulbar and peribulbar blocks?

Answer

Needle-induced globe perforation, particularly in patients with high myopia and an elongated axial globe length, which can cause vitreous hemorrhage, retinal detachment, and vision loss.

#443 • Clinical Subspecialties

Is succinylcholine considered safe to use in a patient with an open globe injury requiring emergency surgery?

Answer

Yes. Succinylcholine does raise intraocular pressure, but current evidence shows this rise is modest and transient, smaller than the pressure rise from coughing or an inadequately controlled intubation. Succinylcholine remains an acceptable rapid sequence induction choice when rapid, reliable airway control is the priority.

#444 • Clinical Subspecialties

What anesthetic goals guide induction in a patient with an open globe injury who also has a full stomach?

Answer

Balance the competing risks of aspiration and elevated intraocular pressure: use a rapid sequence induction technique, with either succinylcholine or a non-depolarizing agent such as high-dose rocuronium as an acceptable alternative, ensure deep anesthesia before laryngoscopy, and avoid coughing, straining, or increases in venous pressure that raise IOP.

#445 · Clinical Subspecialties

What maneuvers other than avoiding succinylcholine help minimize intraocular pressure elevation during airway management for open globe repair?

Answer

Adequate depth of anesthesia before laryngoscopy, avoiding coughing or bucking, gentle atraumatic laryngoscopy, and preventing hypertension, hypercapnia, or venous congestion, all of which can transiently raise IOP.

#446 · Clinical Subspecialties

What is the physiologic effect of pneumatic tourniquet inflation on systemic blood pressure over time?

Answer

Prolonged tourniquet inflation, typically after 30 to 60 minutes, causes a gradual rise in systemic blood pressure known as tourniquet pain or tourniquet hypertension, mediated by ischemic pain from unmyelinated C fibers that is poorly blocked by regional or general anesthesia alone.

#447 • Clinical Subspecialties

What hemodynamic and metabolic changes occur immediately after pneumatic tourniquet deflation?

Answer

Release of accumulated lactate, CO₂, and potassium from the ischemic limb into the circulation transiently decreases blood pressure and core temperature, increases end-tidal CO₂, and can cause transient metabolic acidosis.

#448 • Clinical Subspecialties

What is the recommended maximum safe duration for pneumatic tourniquet inflation on an extremity?

Answer

Generally 2 hours, with periodic deflation for reperfusion (commonly 10 to 15 minutes after each 1.5 to 2 hour period) recommended to reduce risk of nerve injury, muscle ischemia, and rhabdomyolysis.

#449 · Clinical Subspecialties

What is the classic triad of fat embolism syndrome?

Answer

Hypoxemia, neurologic changes (confusion or altered consciousness), and a petechial rash, typically appearing 24 to 72 hours after long bone or pelvic fracture or intramedullary reaming.

#450 · Clinical Subspecialties

What is the management approach for fat embolism syndrome?

Answer

Supportive care including supplemental oxygen or mechanical ventilation for hypoxemia, hemodynamic support, and early fracture stabilization; there is no specific pharmacologic reversal, and corticosteroids remain of unproven benefit.

#451 · Clinical Subspecialties

What causes bone cement implantation syndrome during cemented arthroplasty?

Answer

Pressurization of the medullary canal during cement and prosthesis insertion forces fat, marrow, air, and cement monomer into the venous circulation, causing pulmonary embolization, vasodilation from methylmethacrylate monomer, and activation of complement and thrombin pathways.

#452 · Clinical Subspecialties

What are the clinical features of severe bone cement implantation syndrome?

Answer

Sudden hypoxemia, hypotension, cardiac arrhythmias, increased pulmonary vascular resistance, and in severe cases cardiac arrest, occurring during cement insertion, prosthesis insertion, or joint reduction.

#453 · Clinical Subspecialties

What intraoperative measures reduce the risk of severe bone cement implantation syndrome during hip arthroplasty?

Answer

Communicating with the surgeon before cementing, ensuring adequate intravascular volume and increased FiO₂ before cement insertion, using a cement gun with a distal medullary vent or lavage/suction to reduce intramedullary pressure, and being prepared to treat hypotension immediately.

#454 · Clinical Subspecialties

What is the definition of massive transfusion in adult trauma resuscitation?

Answer

Transfusion of 10 or more units of packed red blood cells within 24 hours, or replacement of a patient's estimated total blood volume within 24 hours, or transfusion of more than 4 units of red cells within 1 hour with anticipated continued need.

#455 · Clinical Subspecialties

What is the target ratio of blood products in a massive transfusion protocol for damage control resuscitation?

Answer

A balanced ratio approximating 1:1:1 of packed red blood cells to fresh frozen plasma to platelets, which mimics whole blood composition and has been associated with improved survival in severely injured, actively bleeding trauma patients.

#456 · Clinical Subspecialties

What is the lethal triad in trauma resuscitation and why is it dangerous?

Answer

Hypothermia, acidosis, and coagulopathy, which form a self-perpetuating cycle: acidosis and hypothermia impair clotting factor function and platelet activity, worsening bleeding, which in turn worsens hypothermia and acidosis, driving mortality upward.

#457 · Clinical Subspecialties

What is the rationale behind permissive hypotension in early trauma resuscitation before hemorrhage control?

Answer

Targeting a lower than normal blood pressure (commonly a systolic pressure around 80 to 90 mmHg in the absence of traumatic brain injury) avoids dislodging early clot formation and dilutional coagulopathy from aggressive crystalloid administration until definitive surgical hemostasis is achieved.

#458 · Clinical Subspecialties

Why is permissive hypotension avoided in trauma patients with concomitant traumatic brain injury?

Answer

Cerebral perfusion pressure depends on adequate mean arterial pressure, and even brief episodes of hypotension are independently associated with worse neurologic outcomes in TBI, so blood pressure should be maintained at normal or higher targets in these patients.

#459 · Clinical Subspecialties

What are the key ventilator management goals in a patient with severe traumatic brain injury?

Answer

Maintain normocapnia (avoiding both hypercapnia, which raises intracranial pressure, and prophylactic hyperventilation, which causes cerebral vasoconstriction and ischemia), ensure adequate oxygenation, and avoid hypotension to preserve cerebral perfusion pressure.

#460 · Clinical Subspecialties

When is therapeutic hyperventilation appropriate in traumatic brain injury management?

Answer

Only as a temporary measure for acute signs of herniation or acutely elevated intracranial pressure refractory to other measures, since sustained hyperventilation causes cerebral vasoconstriction that can worsen ischemia.

#461 • Clinical Subspecialties

Why is rapid sequence induction with cricoid pressure the standard approach for airway management in trauma patients?

Answer

Trauma patients are considered to have a full stomach regardless of last oral intake due to delayed gastric emptying from pain, stress, and opioids, so RSI minimizes the time between loss of protective airway reflexes and securing the airway to reduce aspiration risk.

#462 • Clinical Subspecialties

Which induction agent is generally preferred for hemodynamically unstable trauma patients requiring RSI, and why?

Answer

Etomidate or ketamine are often preferred because they provide relative hemodynamic stability compared with propofol, which can cause significant hypotension in hypovolemic patients; ketamine also provides sympathomimetic support, though etomidate carries a risk of adrenal suppression.

#463 • Clinical Subspecialties

What manual in-line cervical spine stabilization consideration applies during RSI of a trauma patient with a suspected cervical spine injury?

Answer

Manual in-line stabilization should be maintained during laryngoscopy without applying axial traction, and the anterior portion of a rigid cervical collar can be removed to facilitate mouth opening while an assistant stabilizes the neck.

#464 • Clinical Subspecialties

What patient factors favor selection for ambulatory (outpatient) surgery according to standard preoperative screening criteria?

Answer

ASA physical status I to III with medically optimized and stable comorbidities, absence of significant untreated sleep apnea concerns without a plan, adequate social support for postoperative care, and a procedure with an expected short duration and low risk of major complications requiring inpatient monitoring.

#465 • Clinical Subspecialties

What are the components of the simplified Apfel score for predicting postoperative nausea and vomiting risk?

Answer

Female sex, history of motion sickness or prior PONV, nonsmoking status, and use of postoperative opioids; each factor present increases risk, with 0, 1, 2, 3, or 4 factors corresponding to roughly 10%, 20%, 40%, 60%, and 80% risk respectively.

#466 • Clinical Subspecialties

What is the recommended PONV prophylaxis strategy for a patient with an Apfel score of 3 or higher?

Answer

Multimodal prophylaxis combining at least two agents with different mechanisms, such as a 5-HT₃ antagonist (ondansetron) plus dexamethasone, with consideration of a total intravenous anesthetic technique using propofol and avoidance of volatile agents and nitrous oxide.

#467 • Clinical Subspecialties

What are the standard discharge criteria for a patient after ambulatory surgery under general anesthesia?

Answer

Stable vital signs, return to baseline mental status, adequate pain control on oral analgesics, absence of active nausea or vomiting, ability to ambulate and tolerate oral fluids per institutional protocol, and a responsible adult available to accompany the patient home.

#468 • Clinical Subspecialties

Why is voiding no longer a universal requirement for discharge after ambulatory surgery in low-risk patients?

Answer

Routine mandatory voiding prolongs recovery room stay without improving outcomes in patients at low risk of urinary retention, so it is now reserved for patients undergoing procedures with higher retention risk, such as spinal or epidural anesthesia, hernia repair, or urologic surgery.

#469 · Clinical Subspecialties

What is the rationale for multimodal opioid-sparing analgesia in ambulatory surgery?

Answer

Combining acetaminophen, NSAIDs, regional or local anesthetic techniques, and adjuncts like gabapentinoids targets multiple pain pathways simultaneously, reducing total opioid requirements and thereby lowering the risk of opioid-related side effects such as sedation, nausea, and respiratory depression that could delay discharge.

#470 · Clinical Subspecialties

How do peripheral nerve blocks contribute to same-day discharge readiness after ambulatory orthopedic surgery?

Answer

Single-shot or continuous peripheral nerve blocks provide dense site-specific analgesia that reduces opioid consumption and associated side effects, allowing earlier mobilization and satisfaction of discharge criteria, though patients must be counseled about fall risk and unprotected sensation in the blocked limb.

#471 · Special Problems in Anesthesiology

What physiologic response occurs immediately after the electrical stimulus in ECT, and how long does it typically last?

Answer

A brief parasympathetic surge causing bradycardia, hypotension, and sometimes asystole occurs immediately, lasting 10 to 15 seconds, followed by a sympathetic surge with tachycardia and hypertension lasting several minutes.

#472 · Special Problems in Anesthesiology

Why is methohexital historically preferred as an induction agent for ECT?

Answer

Methohexital has minimal anticonvulsant effect at standard induction doses compared to propofol or benzodiazepines, allowing an adequate seizure duration while still providing rapid, brief sedation.

#473 · Special Problems in Anesthesiology

How does propofol affect ECT seizure duration compared to methohexital?

Answer

Propofol significantly shortens seizure duration due to its anticonvulsant properties, which can be useful in patients prone to prolonged seizures but may require stimulus adjustment to achieve therapeutic seizure length.

#474 · Special Problems in Anesthesiology

What is the role of a short-acting muscle relaxant such as succinylcholine during ECT?

Answer

Succinylcholine attenuates the peripheral motor manifestations of the seizure to prevent musculoskeletal injury and fractures without abolishing the central seizure activity needed for therapeutic effect.

#475 · Special Problems in Anesthesiology

Which absolute or relative contraindications should be assessed before ECT?

Answer

Recent myocardial infarction, unstable cardiac arrhythmias, intracranial mass lesions with elevated intracranial pressure, recent stroke, and unstable aneurysms all increase periprocedural risk and warrant careful cardiology or neurology evaluation.

#476 · Special Problems in Anesthesiology

Why might a beta blocker such as esmolol or labetalol be given before ECT?

Answer

It blunts the sympathetic surge in heart rate and blood pressure that follows the seizure, reducing the risk of arrhythmia or hypertensive complications in patients with cardiovascular disease.

#477 · Special Problems in Anesthesiology

How does hyperventilation before stimulus delivery affect ECT seizure quality?

Answer

Mild hyperventilation lowers arterial carbon dioxide, which increases cortical excitability and helps produce a longer, more robust generalized seizure, particularly useful in elderly patients with a high seizure threshold.

#478 · Special Problems in Anesthesiology

What neurologic criteria must be satisfied to declare brain death?

Answer

Coma with a known irreversible cause, absence of brainstem reflexes, and absence of respiratory drive on a formal apnea test, in the absence of confounders such as hypothermia, severe metabolic derangement, or sedative drug effect.

#479 · Special Problems in Anesthesiology

What hormonal changes commonly develop after brain death and contribute to organ dysfunction?

Answer

Loss of hypothalamic-pituitary function causes central diabetes insipidus, decreased thyroid hormone and cortisol levels, and depletion of catecholamine stores, together producing the autonomic storm followed by hemodynamic instability.

#480 · Special Problems in Anesthesiology

What is the standard hormonal resuscitation regimen used in brain-dead organ donors?

Answer

A combination of vasopressin for diabetes insipidus and vasomotor support, intravenous corticosteroids, and thyroid hormone replacement, along with insulin for glycemic control, to stabilize hemodynamics and improve organ yield.

#481 · Special Problems in Anesthesiology

How does anesthetic management differ for donation after circulatory death compared to donation after brain death?

Answer

In donation after circulatory death the patient is not brain dead, so life support is withdrawn in a controlled setting, death is declared after cardiopulmonary criteria, and organ procurement proceeds only after a mandatory no-touch period to confirm irreversibility.

#482 · Special Problems in Anesthesiology

What is the purpose of the mandatory observation period after asystole in donation after circulatory death?

Answer

It ensures autoresuscitation does not occur and confirms the cessation of circulation is permanent before death is legally declared and organ recovery begins.

#483 · Special Problems in Anesthesiology

Why is diabetes insipidus common after brain death, and what is the anesthetic consequence if untreated?

Answer

Destruction of the posterior pituitary eliminates antidiuretic hormone secretion, causing massive free water diuresis that leads to hypernatremia, hypovolemia, and hemodynamic instability unless treated with desmopressin or vasopressin infusion.

#484 · Special Problems in Anesthesiology

What are the core requirements for a safe MRI environment that anesthesiologists must confirm before every case?

Answer

All personnel and equipment entering the scanner room must be MRI-conditional or MRI-safe, ferromagnetic objects must be excluded, and patients must be screened for pacemakers, implants, and other metallic devices that could become projectiles or heat rapidly.

#485 · Special Problems in Anesthesiology

What unique physiologic monitoring hazard exists during MRI, and how is it mitigated?

Answer

Standard ECG cables can act as antennae and cause radiofrequency-induced burns from looped or coiled wires, so MRI-conditional monitoring leads must be used, kept straight, and positioned away from the patient's skin where possible.

#486 · Special Problems in Anesthesiology

What is the greatest safety challenge providing anesthesia in remote or non-operating room locations such as interventional radiology?

Answer

Limited physical access to the patient during the procedure, unfamiliar equipment and layout, inconsistent backup supplies, and greater distance from help make rapid response to an airway or hemodynamic emergency significantly more difficult.

#487 · Special Problems in Anesthesiology

Why do interventional radiology procedures using iodinated contrast carry a special anesthetic risk?

Answer

Contrast media can trigger anaphylactoid reactions, contrast-induced nephropathy, and volume shifts from osmotic diuresis, requiring premedication in high-risk patients, adequate hydration, and vigilance for hypotension or bronchospasm.

#488 · Special Problems in Anesthesiology

What defines decision-making capacity in the context of informed consent for anesthesia?

Answer

The patient must be able to understand the relevant information, appreciate the situation and its consequences, reason about treatment options, and communicate a clear choice; capacity is decision-specific and can fluctuate over time.

#489 · Special Problems in Anesthesiology

How should a preexisting DNR order be handled when a patient presents for surgery?

Answer

The DNR order should not be automatically suspended; instead the anesthesiologist should hold a required reconsideration discussion with the patient or surrogate to clarify goals of care and document a specific perioperative plan for resuscitation.

#490 · Special Problems in Anesthesiology

What are the two main approaches to documenting a perioperative resuscitation plan for a patient with a DNR order?

Answer

A procedure-directed approach specifies which interventions are acceptable such as limited airway management or defibrillation, while a goal-directed approach allows the anesthesia team full latitude to treat any reversible, anesthesia-related event while still honoring the patient's overall goals.

#491 · Special Problems in Anesthesiology

If a patient lacks decision-making capacity and has no advance directive, who typically provides consent for anesthesia?

Answer

A legally authorized surrogate decision maker, usually determined by state or institutional hierarchy such as a spouse, adult child, or other next of kin, provides substituted judgment based on the patient's known wishes or best interests.

#492 · Special Problems in Anesthesiology

What ethical principle is violated if an anesthesiologist proceeds with a case despite a patient's informed refusal of a specific intervention?

Answer

Autonomy is violated, since a capacitated adult has the right to accept or refuse any medical intervention, including resuscitation, even when the refusal conflicts with the clinician's own judgment.

#493 · Special Problems in Anesthesiology

What is the purpose of a root cause analysis after a perioperative adverse event?

Answer

It systematically identifies underlying system and process failures that contributed to the event, rather than assigning individual blame, in order to design corrective actions that prevent recurrence.

#494 · Special Problems in Anesthesiology

What does the Swiss cheese model illustrate about how medical errors occur?

Answer

Each layer of safety defense has inherent gaps or weaknesses, and a serious adverse event occurs only when the holes in multiple independent layers align, allowing a hazard to pass through unchecked.

#495 · Special Problems in Anesthesiology

What are the three phases of the WHO Surgical Safety Checklist?

Answer

Sign-in before induction of anesthesia, time-out before skin incision, and sign-out before the patient leaves the operating room, each involving verbal confirmation of key safety items by the full team.

#496 · Special Problems in Anesthesiology

What is confirmed during the time-out phase of the WHO surgical safety checklist?

Answer

The entire team verbally confirms patient identity, surgical site and procedure, that antibiotic prophylaxis has been given if indicated, and reviews anticipated critical events and equipment concerns before incision.

#497 · Special Problems in Anesthesiology

What is implicit bias, and how can it affect anesthetic care?

Answer

Implicit bias is an unconscious attitude or stereotype that influences judgment and behavior without awareness, and in clinical practice it can contribute to disparities in pain assessment, treatment decisions, and communication with patients from marginalized groups.

#498 · Special Problems in Anesthesiology

Give an example of a documented healthcare disparity relevant to perioperative pain management.

Answer

Studies have shown that Black patients are less likely than white patients to receive adequate analgesia for comparable pain complaints, reflecting both implicit bias and systemic inequities in access to pain treatment.

#499 · Special Problems in Anesthesiology

What structural factors contribute to disparities in access to anesthesia and surgical care?

Answer

Geographic maldistribution of specialists, lack of insurance or underinsurance, language barriers, and lower rates of referral to specialty or subspecialty surgical care all reduce timely access for underserved populations.

#500 · Special Problems in Anesthesiology

What is a key strategy quality improvement programs use to reduce the impact of implicit bias on patient care?

Answer

Standardizing clinical protocols and decision pathways, such as objective pain scoring and structured triage criteria, reduces reliance on subjective judgment where implicit bias is most likely to influence decisions.